

Rainfall Shocks and Time Allocation: Evidence from Bangladesh

Rida Hameed¹

Abstract

In low- and middle-income countries, natural shocks often exacerbate gender disparities in time use. This paper analyzes the impact of rainfall shocks on time allocation in rural Bangladesh. Leveraging nationally representative panel data combined with high resolution satellite rainfall data, I explore how changes in rainfall may influence shifts in time use patterns. I ask whether men and women in rural Bangladesh reallocated time across market work, unpaid care, housework and leisure in response to rainfall shocks, and whether these responses differ by season. Exploiting exogenous variation in monsoon and nonmonsoon rainfall shocks, I find that a 10-percentage point increase in negative monsoon rainfall shocks increases women's unpaid care by 7 minutes per day and decreases her time spent in leisure activities by 24 minutes per day. This can be explained by a rise in caregiving demand. A monsoon deficit increases illness among both children and the elderly thereby increasing the time women spent in unpaid care work. A dry season deficit decreases both unpaid care work by 8 minutes and housework by 17 minutes per day respectively while time spent in leisure activities increases by 27 minutes per day. Men reduce market work by nearly 15 minutes per day due to positive monsoon shocks. Together, these findings shed light on the gendered response to climate shocks and point to exacerbating extant gender gaps in the labor market due to such shocks.

Keywords: rainfall shocks, time use, Bangladesh

JEL classifications: J22, O13, O15, Q54

1 Introduction

In recent years, the frequency and intensity of extreme weather events has increased rapidly due to global warming (Schiermeier 2018; Bohorquez-Penuela and Otero-Cortes 2020; Branco and Feres 2021). The impact is more pronounced in developing countries (Diffenbaugh et al. 2018; Chuang 2019; Bernal and Vlaicu 2023), among vulnerable populations (Flatø, Muttarak, and Pelsier 2017) due to food insecurity (Dimitrova 2021), income losses (Emran and Shilpi 2018; Vogl et al. 2019) and migration as well as in households located in rural areas in developing countries (Bohorquez-Penuela and Otero-Cortes 2020; Patz et al. 2005). In the case of the latter, where rainfed agriculture is heavily relied upon, rainfall uncertainty can lead to income shocks which result in fundamental changes in individual time use patterns. Understanding these dynamics is essential to determine the labor market consequences of climate variability and for designing policies that ensure a gender equal adaptation to climate change.

Prior literature has determined the impact of rainfall shocks on child labor and education (Jensen 2000; Carrillo 2020; Colmer 2021; Aguilar and Vicarelli 2022; Ponnusamy 2026), child health (Tiwari, Jakoby and Skoufias 2016; Abiona 2017; Le and Nguyen 2022; Farris et al. 2023), marriage market outcomes (Corno, Hildebrandt, and Voena. 2020; Trinh and Zhang 2021) income and consumption (Gao and Mills 2018; Carpena 2019) and adult labor supply (Connolly 2008; Gray and Mueller 2011; Afridi, Mahajan and Sangwan 2022; Maitra and Tagat 2024; Diallo and Ondo 2025). Research on the impact of weather shocks on time allocation is primarily focused on developed countries (Kruger and Neugart 2018; Nguyen et al. 2021), with limited research on developing countries (Nasuda, Kanno, and Sakurai 2011; Namhata et al. 2025). This paper aims to determine the impact of rainfall shocks on time allocation in rural Bangladesh. Specifically, I want to answer whether and how do time allocation responses to rainfall shocks differ by gender and season. Examining these impacts can help shed light on how the burden of climate shocks is borne by both men and women in the household.

Having a wet, humid and warm climate, Bangladesh is considered as one of the most vulnerable countries to climate change and extreme weather events such as floods (Islam et al. 2014; Dastagir 2015; Ahmed et al. 2021; Namhata et al. 2025). The monsoon season (June to October) in the country is considered as the main driver of agricultural output, with the country being ranked as the third highest rice producing country in the world (Al Mamun et al. 2021). Furthermore, gender inequality is amplified in time use activities with women spending more time than men in both market and unpaid care work (Anxo et al. 2011; Ferrant, Pesando and Nowacka 2014) and are more time poor (Solotaroff et al. 2019). This makes Bangladesh an ideal setting to examine the relationship between climate shocks and intra household labor allocation decisions.

This paper aims to investigate how rainfall shocks impact time allocation in rural areas of Bangladesh. More specifically, I try to answer the following question: do men and women allocate time differently in market work, unpaid care, housework and leisure activities in response to rainfall shocks? To answer this question, I match high resolution daily precipitation satellite data taken from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) with rural household panel data from Bangladesh Integrated Household Survey (BIHS) (2011-12, 2015, 2018-19) having information time spent in market, housework, unpaid care and leisure activities. Much of the prior literature uses local weather station data to measure rainfall shocks. This faces the limitation of having missing data points and has coarse geographic coverage (Agamile and Lawson 2021). To counter the issues prevalent in local weather station data, I rely on satellite rainfall data, CHIRPS, available at a resolution of 5 km daily. Utilizing exogenous variation in monsoon and nonmonsoon rainfall shocks, I adopt an OLS fixed effects estimation to determine the impact of rainfall shocks on time use patterns. Specifically, I investigate how the change in time allocation varies between males and female members of the household as rainfall shocks are often gendered (Maitra and Tagat 2024). The identification in this paper relies on rainfall variation within individuals across survey rounds in rural Bangladesh. Because weather is not randomly

assigned across space, estimated associations between rainfall and time use may be due to factors other than weather. In order to address this, I define shocks as deviations from long term rainfall average. This helps to isolate responses to unanticipated rainfall at a given moment in time and space (Deschenes and Greenstone 2007; Burke and Emerick 2016; Sestero, Ricke-Gilbert and Cook 2018) while the inclusion of individual fixed effects help absorb all time-invariant individual and geographic characteristics including long-run climate and residential sorting or households’ adaptation to historical rainfall risk (Colmer 2021). The second concern is that of temperature which is a time-varying correlate of rainfall also independently affects time allocation activities (Graff Zivin and Neidell 2014; Dell, Jones and Olken 2014). Therefore, I control for it in my specification. Conditional on individual, wave by division, day of week and month of interview fixed effects, the identifying assumption is that rainfall deviations are orthogonal to other time-varying determinants of time use.

Empirical results suggest that an increase in negative monsoon rainfall shocks is associated with an increase in women’s unpaid care by 7 minutes per day and a reduction in the time they spend in leisure and other activities by 24 minutes per day.² I find that a monsoon rainfall deficit raises the demand for caregiving by increasing the number of ill dependents in the household, including higher fever among children under 5 and illness among the elderly. On the other hand, men do not adjust their time use significantly in response to a negative monsoon shock.

A dry season rainfall deficit is associated with women’s housework and unpaid care decreasing by 17 and 8 minutes respectively with their leisure time increasing by 27 minutes per day. Furthermore, there is no impact on the production of Boro rice, the dry season crop grown mainly under irrigation between November and May. The nonmonsoon period precedes the monsoon shock and is thus measured nearly a year before responses in the time-

²Each of the four time-use categories—market work, unpaid care, housework, and leisure—is estimated in a separate regression, and together they exhaust the respondent’s day. A fall in time spent in leisure is offset by an increase in unpaid care together with small increases in housework and market work that are not statistically distinguishable from zero, so the estimated changes across categories sum to approximately zero.

use module. Because of this, I do not attribute a causal mechanism to the non-monsoon estimates. For men, a 10 percentage point increase in positive rainfall monsoon shock reduces time spent in market work by 15 minutes per day. Findings from this paper reveal that women reallocate time between home production and leisure due to negative monsoon shocks while men’s time use remains unchanged. Positive monsoon shocks, however, reduce the time men spend in market work activities with no corresponding change observed in women’s time use patterns.

This paper contributes to the existing literature in three main ways. First, it provides evidence on the impact of smaller, more frequent rainfall shocks which are rarely offset by aid (Pham 2021) on time allocation dynamics. Second, it sheds light on gendered responses to climate shocks (Afridi, Mahajan and Sangwan 2022) across time allocation activities including market work, unpaid care, housework and leisure activities instead of labor supply alone. Third, it adds to the current literature on climate shocks and time allocation in developing countries particularly in South Asia by combining rich rural panel survey data with high resolution satellite rainfall to answer this research question.

The paper proceeds as follows: Section 2 focuses on the context and background; Section 3 presents the conceptual framework and Section 4 sheds light on previous literature. Section 5 focuses on data and descriptive statistics while Section 6 discusses the empirical strategy adopted in this paper. Section 7 reports and discusses the results and Section 8 concludes with discussion.

2 Context

2.1 Rainfall in Bangladesh

According to the World Risk Index 2023, Bangladesh ranks 9th in the world vulnerable to climate disaster risk (Frege et al. 2023). From 2000-19, the country experienced 185 extreme weather events including floods, tropical cyclones, rising sea levels, heavy precipitation and

extreme heat and drought events (Huq et al. 2024). These events lead to increased mortality, destruction of infrastructure, crops and livestock, an increase in migration and livelihood as well as loss of income (Brown et al. 2013; Damania et al. 2020; Frame et al. 2020; Newman and Noy 2023). The climate scenario in Bangladesh is characterized by wet, humid and warm days. The monsoon season lasts from June to October and plays a crucial role in agricultural productivity and household income in the country.

Figure 1 plots the total weekly rainfall experienced in Bangladesh between 2012 and 2019 highlighting inter-annual variation across the years. Rainfall follows a distinct seasonal pattern, with low precipitation during winter months (November–February) and sharp peaks during the monsoon season (June–October).

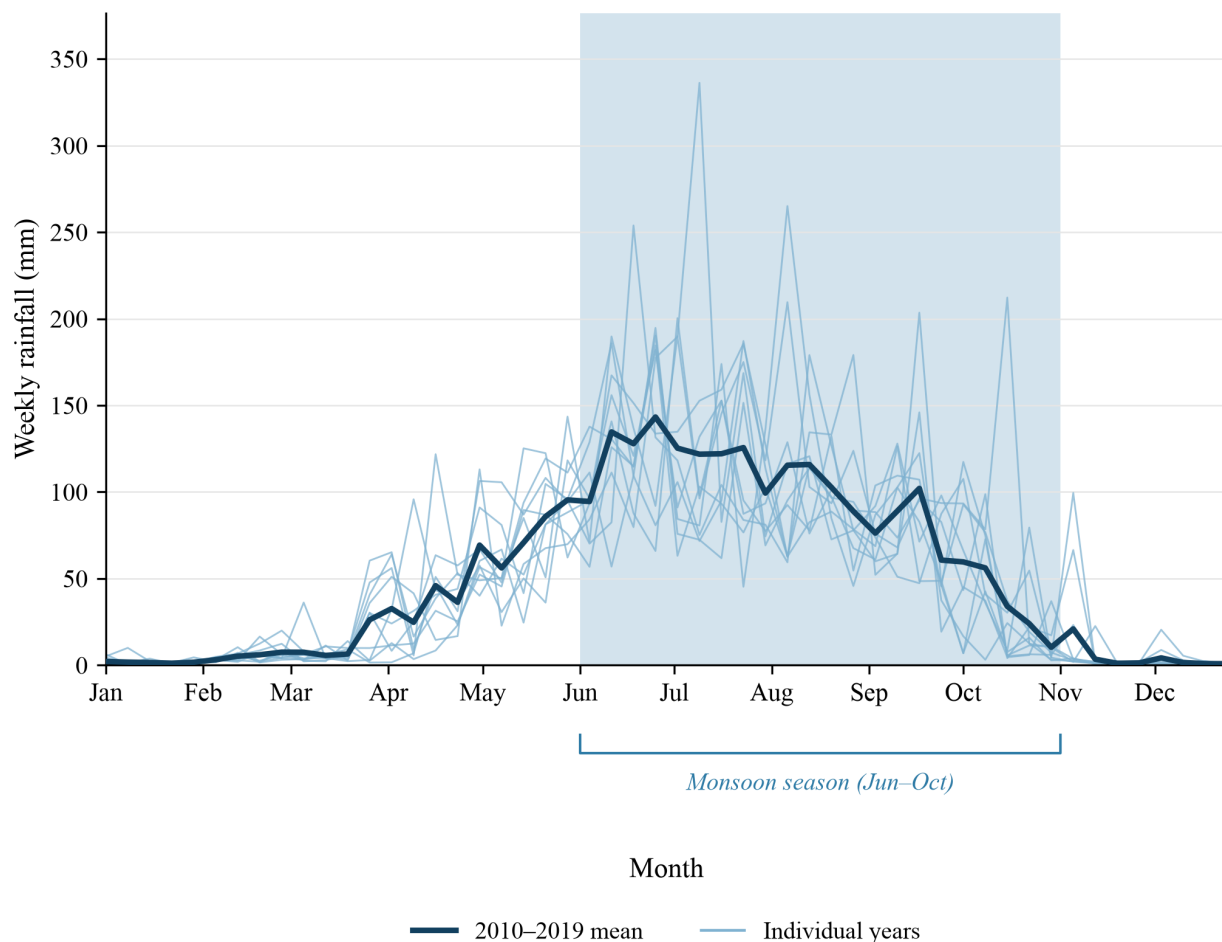


Figure 1: Weekly rainfall in Bangladesh, 2010–2019.

Notes: Weekly rainfall is total precipitation within each week of the year, computed from CHIRPS daily precipitation averaged over all grid cells covering Bangladesh. Thin lines denote individual years; the bold line is the 2010–2019 mean. The shaded band marks the monsoon season (June–October).

2.2 Rainfall and Time Allocation in Bangladesh

Men and women allocate their time differently across the day. While men focus on spending more time in market work activities as found in cross-country evidence (Apps 2004; Ferrant, Pesando and Nowacka 2014), in the United States, women specialize in housework and unpaid care activities and also spend more aggregate time in all activities combined, making them time poor (Sayer 2005). Women carry 3/4th of the burden of unpaid care work (Addati et al. 2018; Charmes 2019). Unpaid services, if remunerated, would represent 9% of global GDP (Addati et al. 2018), with this number being as high as 20% of GDP, making its value

greater than that of other large sectors including manufacturing and commerce in Colombia (Torres-Higuera 2024).

Bangladesh exhibits a rigid division of labor deeply embedded in gender norms. In rural areas, men are primarily involved in agricultural production including both preharvest and harvest activities. On the other hand, women are either primarily engaged in domestic care work including performing domestic tasks or caring for household members (Jiao, Li and Liu 2021) or in postharvest activities or activities that keep them close to the homestead for example food processing and preparation and animal husbandry (Vitelloszi and Giannelli 2024). Because of this division of labor, women are more vulnerable to adverse climate shocks and have less effective coping strategies (Islam et al. 2017; Solotaroff et al. 2019). Because women are primarily involved in home production rather than paid employment, they possess fewer productive or market resources (Solotaroff et al. 2019). Furthermore, households substitute home production in response to climate shocks when they are unable to afford market goods which increases unpaid work for women. In this context, climate shocks may further exacerbate such well-defined gender roles and warrant further investigation of the resulting intra-household time allocation.

3 Literature Review

In rural, agrarian economies, rainfall primarily leads to sectoral reallocation of time use activities. Droughts lead to individuals shifting from agricultural to non-agricultural employment, working secondary jobs and also increasing the total time spent in employment in rural India (Rose 2001) and reducing agricultural employment in Brazil (Branco and Feres 2020). Similarly, excessive rainfall increases market work and reallocates labor from on-farm to off-farm and wage employment in rural Zambia (Nasuda, Kanno and Sakurai 2011) and in rural India (Maitra and Tagat 2024). On the other hand, prior evidence also points to positive rainfall shocks increasing school enrollment and reducing market work for young

adults, with this trend reversing in the case of droughts in Madagascar (Marchetta, Sahn and Tiberti 2019). While most of these papers examine gender, they do not examine how the impact of rainfall shocks varies across agricultural seasons.

Prior research also reveals the gender asymmetry present in time allocation responses. Rainy days lead to men shifting their leisure time to time spent in market work activities or increasing time spent in wage employment (Connolly 2008; Connolly 2018; Maitra and Tagat 2024). On the other hand, women’s paid labor supply remains unresponsive to rainfall shocks (Connolly 2018; Jiao, Li and Liu 2021). While rainfall leads to an increase in time spent on economic activities and an increase in leisure time, with gender gaps reducing in paid work due to men reducing their own work time rather than women increasing theirs (Namhata et al. 2025). For women, weather shocks lead to a deepening of time poverty, with women spending the same or more time on unpaid care and housework without a proportionate decrease in market work (Boyd 2023; Garg, Gibson and Sun 2020; Jiao, Li and Liu 2021).

This asymmetric gender response to time allocation can be attributed to two mechanisms: the added worker effect and a relaxation of domestic time constraints. In the case of the former, women reallocate time from home production activities to market work, with men increasing their time spent in domestic work as a result (Canessa and Giannelli 2021; Lee et al. 2021). In the case of domestic work, rainfall shocks have an ambiguous impact on female time allocation. On the one hand, excess rainfall reduces the time women spend on certain activities such as water collection (Boyd 2023; Maitra and Tagat 2024). In the case of floods, the reduction in time spent on domestic chores is also accompanied with an increase in time spent on leisure activities (Vitelloszi and Giannelli 2024). However, excessive rain leading to flooding can contaminate water supplies, thereby increasing the time cost of getting water and also have a detrimental impact on water and sanitation infrastructure. Excessive rain also increases the incidence of non-communicable diseases through waterborne and vector-borne illness such as malaria and dengue and increase caregiving burden for women. Excess rain may also delay treatment and recovery period for the elderly and the chronically ill thereby

increasing the unpaid care time of women (Floro 2026). Because of these reasons, the net effect of such climate shocks on women’s unpaid care work is ambiguous and dependent on which of these forces dominates.

Prior literature has explored the impact of weather shocks on reshaping the allocation of time (Graff Zivin and Neidell 2014; Neidell et al. 2021; Somanathan et al. 2021; Connolly 2018). Extreme temperature leads to lesser time devoted to market activities, with the effect being stronger for workers in high-exposure industries such as agriculture, construction and mining or those located in warmer regions (Graff Zivin and Neidell 2014). Furthermore, extreme heat days also lead to a reallocation of outdoor to indoor leisure due to presence of air conditioning (Zivin and Neidell 2014; Connolly 2018; Neidell et al. 2021). In addition to changing aggregate time, weather shocks also change when, where and by whom activities are performed. Weather shocks lead to intertemporal substitution across days, and towards cooler hours (Connolly 2008; Kruger and Neugart 2018; Cosaert, Castro and Tatsiramos 2023). However, this rescheduling is confined to workers having flexible schedules and fewer family-related constraints (Cosaert, Castro and Tatsiramos 2023).

4 Conceptual Framework

Rural households in Bangladesh face incomplete credit and insurance markets. Because of this, agricultural production shocks also become time allocation shocks. Labor allocation is a household decision making process instead of an individual one (Demurger, Fourmies and Yang 2010; Jonasson and Helfand 2010), with rainfall shocks having both an immediate and long-term persistent impact. Weather shocks can lead to different labor allocation responses. Rural households may engage in more work or substitute between on-farm and non-agricultural activities. They may also engage in secondary, non-agricultural jobs to supplement household income as an autonomous mitigation strategy. More affluent households may also choose to migrate to areas with more favorable weather conditions.

Rainfall shocks impact time allocation patterns through three channels. This first is the productivity or substitution channel. This means that positive rainfall shocks increase marginal product of farm labor and increases the number of workers engaged in on-farm work. However, excessive rainfall may also lead to destruction of crops and reduce both productivity and on-farm labor demand. On the other hand, deficient rainfall pushes labor into off-farm or non-agricultural activities or induces workers to take up secondary non-agricultural jobs or work in multiple jobs, home production or leisure (Rose 2001; Branco and Feres 2020).

The second channel is the income channel. Rainfall deficits decrease household income thereby reducing the consumption of normal goods including leisure. The shock may also lead to an added worker response when income loss threatens household subsistence, where secondary workers including women and children may work in income-generating activities (Canessa and Giannelli 2021; Lee et al. 2021). In this case, deficient rain would increase the time spent in market activities.

The third channel is that of home production. Rainfall shocks impact time use patterns directly by changing the time cost of water and fuel collection or increasing the burden of caregiving when higher rainfall increases the disease burden among household members (Floro 2026). In such an instance, individuals' time spent in domestic chores is impacted even if time spent in market work remains unaffected. Since women bear the brunt of domestic production, climate shocks operating through this channel are inherently gendered.

Gendered constraints play a pivotal role in intra-household allocation due to climate shocks. In Bangladesh, social norms dictate women to contribute to domestic and unpaid care work and also restrict their mobility to engage in off-farm activities (Emran and Shilpi 2018; Afridi, Mahajan and Sangwan 2022). On the other hand, men are less likely to engage in home production and therefore adjust their time allocation between market and non-market activities.

The importance of these channels varies across seasons. In the monsoon season, agri-

cultural production is primarily rainfed. Because of this, both excessive and deficient rain shocks serve as income shocks. On the other hand, crop production in the nonmonsoon season relies primarily on irrigation because of which rainfall shocks do not serve as income shocks. Hence, an identical rainfall shock can lead individuals to respond differently depending on the season in which the shock is experienced.

Within this framework, the empirical analysis tests three predictions. First, rainfall shocks impact members engaged in market work, with a possibility of reallocation between farm and off-farm activities. Second, deficient rain may increase women’s unpaid care and domestic work due to a binding home production channel. This may exacerbate time poverty as market work would still remain largely unaffected. Third, because both the income and substitution impacts operate in opposing directions, the net impact on market work is ambiguous. Deficient rain may lead to a reduction in market work if the substitution channel dominates. On the other hand, if household income falls to threaten subsistence levels, the income effect outweighs the substitution effect and results in an increase in market work. Findings from this paper would therefore help to determine which of these effects is more pervasive in rural developing contexts such as Bangladesh. Understanding these inequalities is imperative to ensure that better policies having a more gender-driven response to climate change adaptation are designed which account for higher vulnerability women are exposed to than their male counterparts (Jost et al. 2016; Rao et al. 2019).

5 Data and Descriptive Statistics

5.1 Bangladesh Integrated Household Survey

The data on time allocation was taken from Bangladesh Integrated Household Survey (BIHS) conducted between 2011 and 2019 (October 2011-March 2012, January-June 2015, November 2018-April 2019) by International Food Policy Research Institute (IFPRI) in Bangladesh. This data set provides information on 6503 households in 325 primary sampling units and

is representative of rural areas in all seven divisions of the country. The survey collects detailed information on demographic characteristics, education, employment, agricultural production, time allocation and women’s empowerment.

Information pertaining to time use activities is determined from the time allocation module of the survey. This module records information on time spent by both men and women on work and leisure activities in the last 24 hours (4:00 am from the previous morning to 3:59 am of current day). Household time use is classified into 4 main categories: market work, unpaid care, housework and leisure and other activities. Market work includes activities such as working as employed, own business work, farming/fishing and weaving, sewing and textile care. Unpaid care work includes care for children and for adults/elderly. Housework includes shopping, cooking and domestic work. Leisure and other activities involve personal well-being and socializing activities including sleeping and resting, eating and drinking, personal care, watching TV or listening to radio, reading, sitting with family, exercising, social activities and practicing hobbies, school (also homework), commuting, travelling, religious activities and other activities.

The working sample is composed of time use diary respondents having complete information on time-use outcomes, weather and household controls. BIHS has information on household GPS coordinates for 6306 of 6503 households. Because these are needed to merge in rainfall and temperature data, households without coordinates are dropped. I also exclude responses from the time allocation module for which the total time spent in activities during the day falls below the 2nd (1290 minutes) or above the 98th percentile (1530 minutes). I correct for and drop implausible age of respondents falling below 12 years or above 95 years of age dropping a total of 7 observations. I further restrict my sample to individuals observed in at least two rounds as identification comes from within individual variation. Because of this, households in which a different member provided responses to the time use diary is dropped. The final sample is comprised of 25176 members, with 15175 women and 10001 men in 5907 households. Figure 2 below plots these surveyed households in the working

sample across the country.

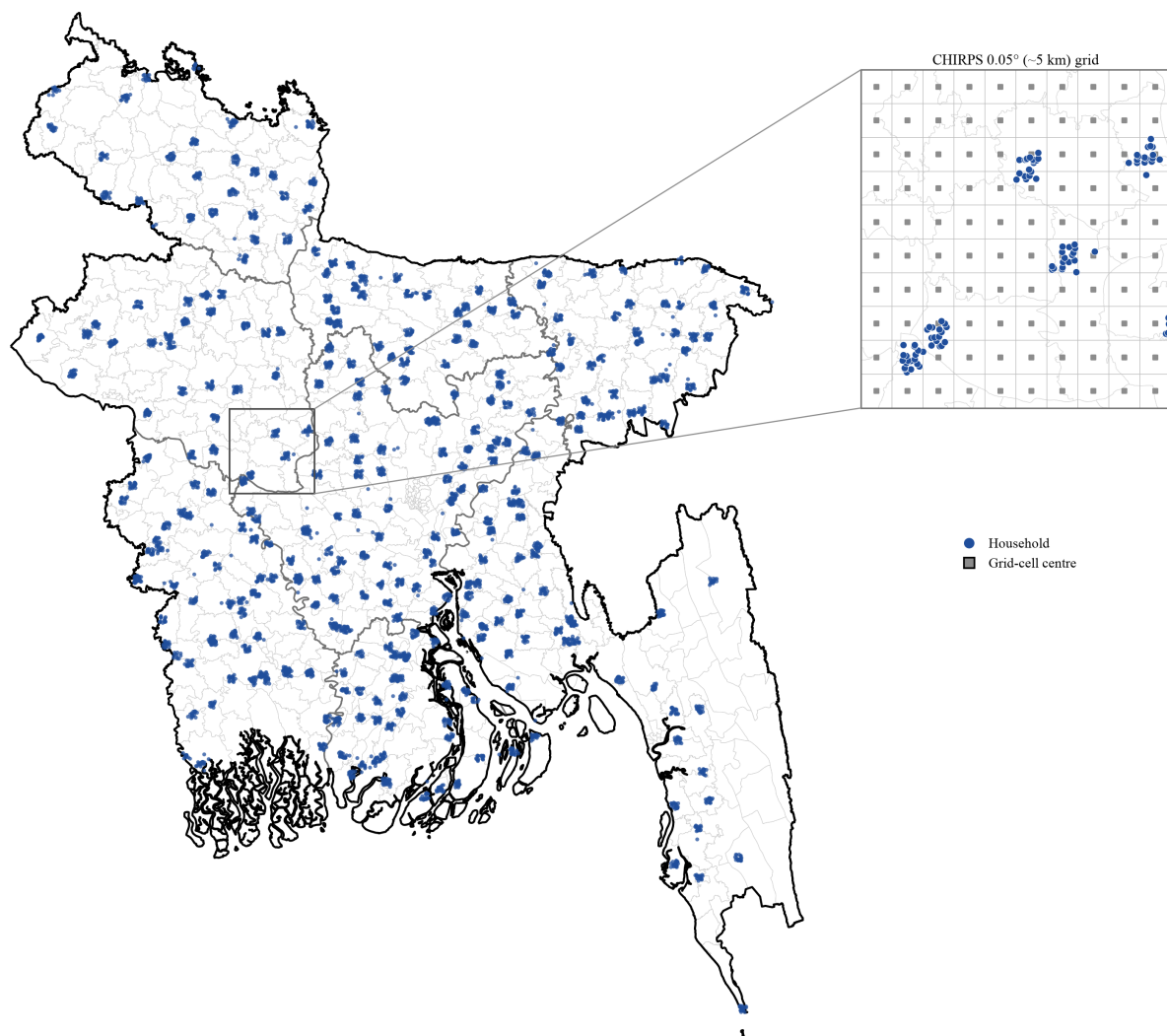


Figure 2: Distribution of BIHS households across Bangladesh.

Notes: Each dot is a BIHS working-sample household ($N = 5,907$) placed at its GPS coordinates. Thin grey lines are upazila boundaries and medium grey lines are the eight administrative divisions. The inset shows the 0.05° (~ 5 km) CHIRPS rainfall grid; each household is matched to its nearest grid cell to construct rainfall shocks.

To determine time allocation in different activities, I use the time allocation module of the survey. This module records information on time spent by both men and women on work and leisure activities in the last 24 hours. Household time use is classified into 4 main categories: market work, unpaid care, housework and leisure and other activities. Table 1

reports the activities included in each of these categories.

Table 1: Categories of Time Allocation

Market Work	Unpaid Work	Housework	Leisure and Other Activities
Work as Employed	Care for children	Shopping/getting service	Sleeping and resting
Own Business Work	Care for adults/elderly	Cooking	Eating and drinking
Farming/Fishing		Domestic Work	Personal Care
Weaving, sewing, textile care			Watching TV/listening to radio
			Reading
			Sitting with Family
			Exercising
			Social Activities
			Practicing hobbies
			School (also homework)
			Commuting
			Travelling
			Religious activities
			Other activities

Figure 3 shows the average time allocated by men and women in a day. While the average time spent on leisure and other activities is nearly the same for both genders (15 hours), men spend around 6.5 hours in market work, while women spend nearly 6 hours in housework. Men, on the other hand, spend around 1.5 hours on housework and an average of around 8 minutes on unpaid care work. Women devote around 1 hour 20 minutes to market work.

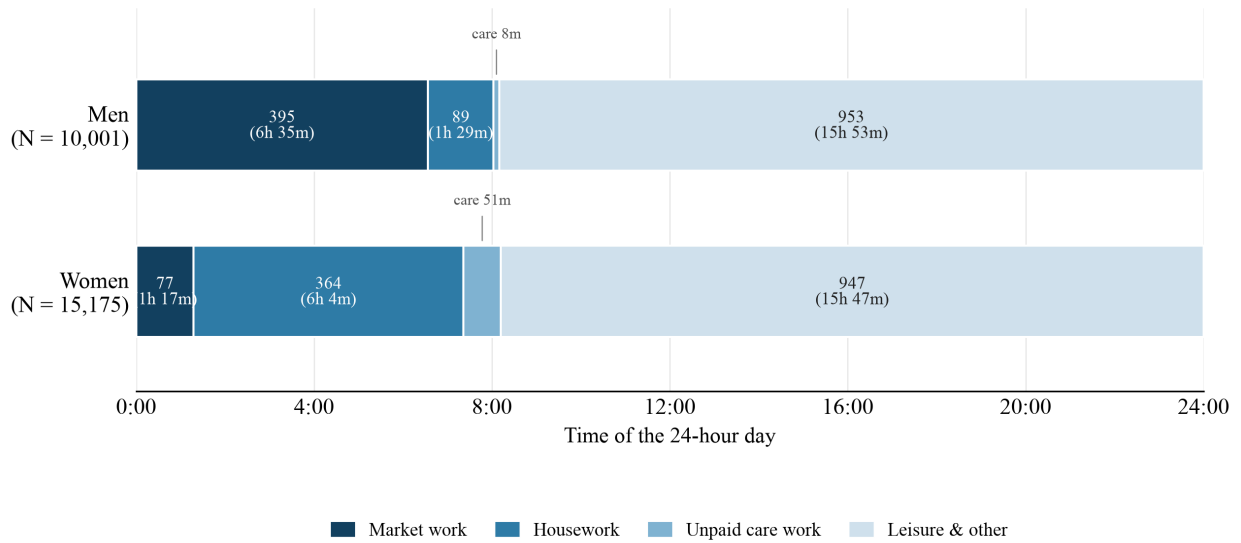


Figure 3: Gendered allocation of the 24-hour day, rural Bangladesh.

Notes: Bars show mean minutes per day that primary male and female BIHS respondents allocate to the four time-use categories.

5.2 Rainfall Data

The rainfall data used in this study is obtained from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS). Developed by the U.S. Geological Survey (USGS) and the Climate Hazards Group at the University of California, Santa Barbara (UCSB), this daily dataset has a spatial resolution of $0.05^\circ \times 0.05^\circ$ (approximately 5km x 5km) and is available from 1981 to near present.³ It combines high resolution satellite data with ground station data to create a spatially gridded daily rainfall dataset.

I extract daily rainfall data and aggregate it at the weekly level for both the monsoon (June to October) and the non-monsoon periods (November to May) for each household using its GPS coordinates to identify its nearest CHIRPS grid to determine rainfall shocks.

⁴Although the Standardized Precipitation Index (SPI) is a standard rainfall metric, I am unable to use it in my setting. This normalizes rainfall data by fitting a gamma distribution and applying an inverse-normal transformation (McKee, Doesken and Kleist 1993). The presence of 0 and near 0 rainfall weeks yield a poor gamma fit and the index fails the standard test of normality. Because of this, I compute the share of normal rainfall index for each household as follows:

$$Share\ of\ Normal_{wht} = \frac{R_{wht}}{R_{wh}} \quad (1)$$

where R_{wht} is the observed rainfall for a given week w at household h in year t , R_{wh} is the long-term mean calculated over 30 years from 1981 to 2011 for week w at household h . This long-term mean precedes the timing of the survey waves to ensure independence of

³The precipitation data integrates infrared Cold Cloud Duration (CCD) observations with station data, using ‘smart’ interpolation in sparsely gauged regions to produce daily, pentadal and monthly estimates with a 2-day preliminary and 3-week final latency (Funk et al. 2015).

⁴For each survey round, the rainfall shocks are constructed from the two seasons preceding the interview: the monsoon (June–October) and the non-monsoon (November–May). The monsoon window is the season immediately preceding the interview, whereas the non-monsoon window falls before that monsoon and is therefore measured roughly a year before the interview. The (non-monsoon, monsoon) windows by round are: BIHS 2011–12 (fieldwork October 2011–March 2012), November 2010–May 2011 and June–October 2011; BIHS 2015 (January–June 2015), November 2013–May 2014 and June–October 2014; and BIHS 2018–19 (November 2018–April 2019), November 2017–May 2018 and June–October 2018 (Figure 4).

rainfall realizations used to construct the shock variables. This index helps to calculate a scale-invariant measure by normalizing weekly rainfall at the household level and determines the average over survey years.

To account for asymmetric effects of rainfall shocks, I decompose the share-of-normal rainfall index into separate variables for negative and positive deviations. Specifically, I define a negative shock as follows:

$$Negative\ Rainfall\ Shock_{wht} = \begin{cases} 1 - Share\ of\ Normal_{wht}, & \text{if } Share\ of\ Normal_{wht} < 1 \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

And a positive shock as below:

$$Positive\ Rainfall\ Shock_{wht} = \begin{cases} Share\ of\ Normal_{wht} - 1, & \text{if } Share\ of\ Normal_{wht} > 1 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Next, I aggregate the weekly deviations to the season by averaging over the weeks within each season, yielding the shock measures that enter the estimation:

$$\begin{aligned} Negative\ Rainfall\ Shock_{sht} &= \frac{1}{|\mathcal{W}_s|} \sum_{w \in \mathcal{W}_s} Negative\ Rainfall\ Shock_{wht}, \\ Positive\ Rainfall\ Shock_{sht} &= \frac{1}{|\mathcal{W}_s|} \sum_{w \in \mathcal{W}_s} Positive\ Rainfall\ Shock_{wht} \end{aligned} \quad (4)$$

where $|\mathcal{W}_s|$ is the number of weeks in season s . The seasonal shock variables, indexed by season s , household h , and year t , are the rainfall shock measures used throughout the analysis.

I measure rainfall shocks as continuous deviations from the long run norm. This allows to capture the full intensity of the shock and uses all variation in the rainfall distribution rather

than using a binary indicator for drought or flood. It also helps to avoid the arbitrariness inherent in defining an extreme climate event, where results may be driven by the threshold chosen. Households exposed to exactly normal rainfall ($Share\ of\ Normal_{sht} = 1$) are treated as the reference group. This decomposition helps determine the extent of household exposure to rainfall shocks, interpreted as the marginal effect of increasingly severe negative or positive shocks relative to the baseline reference group. Once calculated, I spatially merge this data with the household dataset using the geographical location of households.

The construction of rainfall shocks is dependent on distinct cropping cycles in rural Bangladesh. Aman rice, the dominant rainfed crop, is sown during the monsoon period (June-October) and harvested in November-December. In the non-monsoon period spanning from November to May, Boro rice and Rabi crops are planted. This period also includes post-harvest processing of agricultural crops which comprises of time allocated by households to these activities. Figure 4 below plots the timeline followed to construct contemporaneous rainfall shocks relative to the time periods in which the surveys are conducted. For each survey wave, the monsoon period spans over June to October while the non-monsoon period is measured over November to May. Households exposed to exactly normal rainfall are treated as the reference group. This decomposition helps to determine the extent of household exposure to rainfall shocks to be interpreted as the marginal effect of increasingly severe negative or positive shocks relative to the baseline reference group. Once calculated, I spatially merge this data with the household dataset using the geographical location of households.

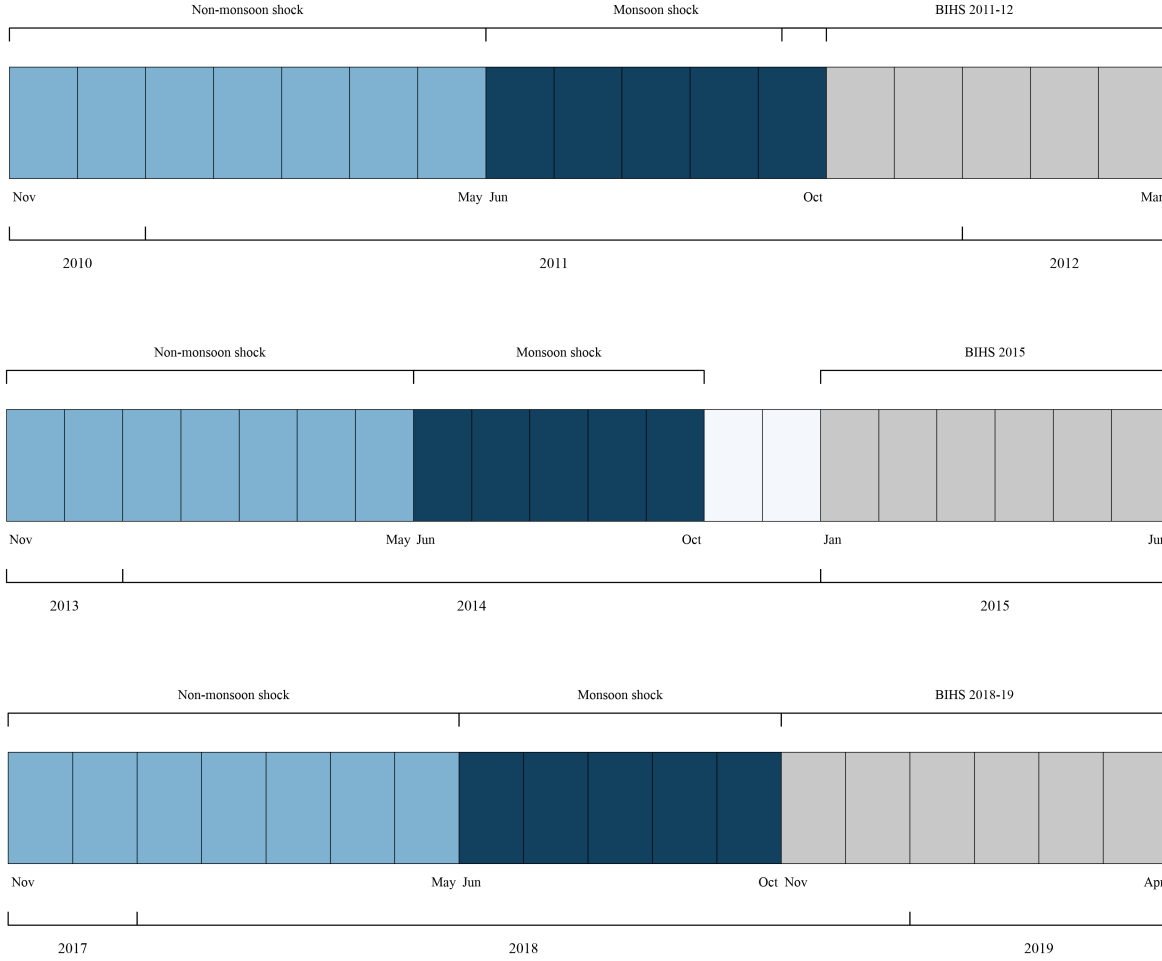


Figure 4: Timeline of current rainfall shocks by season and survey wave.

Notes: Each row corresponds to one BIHS survey wave. Current rainfall shocks are built from the two seasonal windows immediately preceding the wave: the monsoon season (Jun–Oct, dark) and the non-monsoon season (Nov–May, light). The grey block marks the survey fieldwork window.

Figures 5 and 6 plot the distribution of negative and positive rainfall shocks between 2011 and 2019 disaggregated by season (monsoon vs. non-monsoon) and time (current vs. past) respectively. While negative monsoon shocks are concentrated between 0.2 and 0.4, non-monsoon shocks are shifted rightwards and primarily lie between 0.3–0.6. This is indicative of the latter being more severe in magnitude. Positive shocks are plotted in figure 6. While positive current monsoon shocks are primarily concentrated below 0.2, other monsoon and non-monsoon shocks are characterized with higher frequencies of moderate to large rainfall

events.

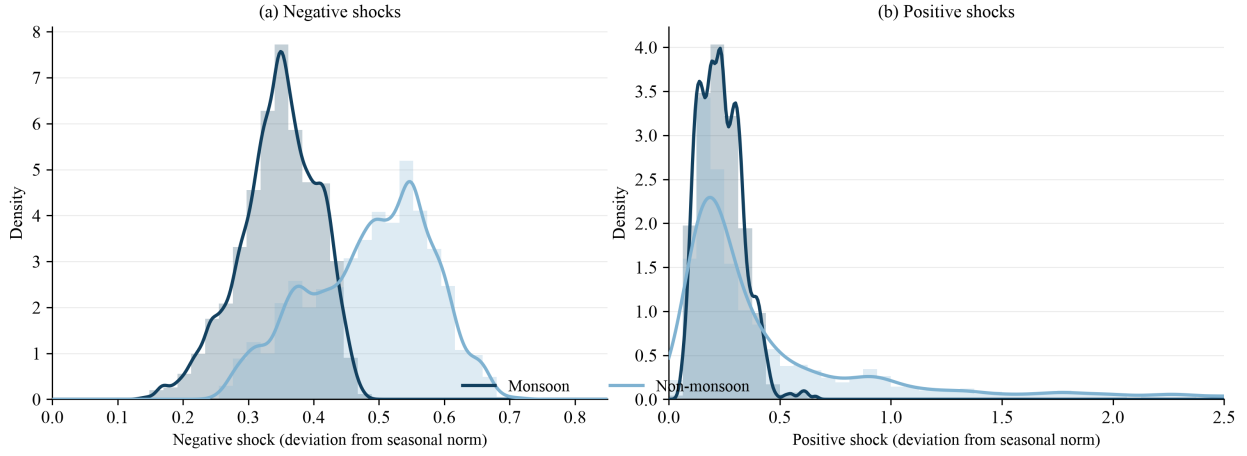


Figure 5: Distribution of current rainfall shocks by season.

Notes: Kernel densities (with underlying histograms) of the household-level rainfall shocks, for monsoon (Jun–Oct) and non-monsoon (Nov–May) seasons.

5.3 Temperature Data

Data on temperature is obtained from European Center for Medium-Range Weather Forecasts (ECMWF) ERA5. This provides hourly data for 2-meter air temperature and is available at a spatial resolution of 0.25 degree grid. I calculate daily means and then average them over the season as with rainfall shocks. Following Abiona (2017), average temperature is determined as:

$$T_{hst} = \frac{T_{hst} - \bar{T}_{hs}}{\sigma_{hs}} \quad (5)$$

where T_{hst} is household h 's average temperature experienced in season s in round t and $\bar{T}_{hs}$ and σ_{hs} are mean and standard deviation of that household's seasonal temperature over the years 1981–2011. As temperature is correlated with rainfall and also independently affects time use decisions, I control for it to isolate rainfall variation from correlated heat (Auffhammer et al. 2013; Graff Zivin and Neidell 2014; Dell, Jones and Olken 2014).

Table 2 presents the descriptive statistics of the key variables used in the analysis. These descriptives help provide a comprehensive overview of the household demographics, rain-

fall shocks and time use patterns prevalent in the data. Panel A provides information on weather variables. Negative rainfall shocks fall about 35–49% short of the long run rainfall experienced by households in the sample while positive rainfall shocks exceed the long-term average by 23–50%. Monsoon temperature averages 1.8 standard deviations above the long-term norm while average nonmonsoon temperature is 0.14 standard deviations below the historical average. Panel B summarizes time use variables from BIHS and sheds light on the gendered division of labor present in the sample. Men work 395 minutes (6.6 hours) on average in market work compared to women working 77 minutes (1.3 hours) on average per day. This pattern reverses in the case of housework and unpaid care, with men spending 89 minutes (1.5 hours) per day in home production and 8 minutes in unpaid care as compared to women allocating 364 minutes (6.1 hours) in housework and 51 minutes in unpaid care. Both genders spend nearly 950 minutes (15.8 hours) in leisure and other activities.

Panel C reports individual and household characteristics. There is an average of nearly 4 members per household with a maximum of 20 members. The asset indexes are created using principal component analysis for durable goods, agricultural equipment and livestock and have a mean of 0 by construction, with positive values indicating above average asset wealth and negative values representing below average asset wealth.

Table 2: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Panel A: Rainfall Shocks					
Negative Current Rainfall Shocks	25,176	.346	.059	.137	.479
Positive Current Rainfall Shocks	25,176	.234	.095	.044	.662
Negative Current Rainfall Shocks (nonmonsoon)	25,176	.485	.093	.244	.720
Positive Current Rainfall Shocks (nonmonsoon)	25,176	.497	.557	.005	3.640
Panel B: Time Allocation (minutes/day)					
Men's Market Work	10,001	394.872	248.747	0	1320
Men's Unpaid Care Work	10,001	7.556	28.407	0	405
Men's Housework	10,001	88.776	135.892	0	885
Men's Leisure	10,001	952.805	229.568	105	1530
Women's Market Work	15,175	77.482	130.715	0	1050
Women's Unpaid Care Work	15,175	50.756	79.551	0	675
Women's Housework	15,175	363.531	151.235	0	1005
Women's Leisure	15,175	946.807	164.829	240	1515
Panel C: Household Characteristics					
Temperature Shock (monsoon)	25,176	1.807	.539	-.141	3.097
Temperature Shock (nonmonsoon)	25,176	-.145	.454	-1.275	1.654
Household Size	25,176	4.345	1.645	1	20
Number of Elderly in Household	25,176	.376	.609	0	4
Number of Children Aged 0–4	25,176	.412	.612	0	5
Number of Children Aged 5–17	25,176	1.332	1.105	0	7
Durable Asset Index	25,176	.138	2.092	-6.325	8.891
Agricultural Asset Index	25,176	.091	1.606	-1.766	12.287
Livestock Index	25,176	.043	1.187	-.882	39.268

Note: Sample size is 25,176 observations in 5,907 households, of which 10,001 observations are men and 15,175 are women. Rainfall shocks are the season's mean weekly proportional deviation from the household's 1981–2011 weekly norm. Asset indexes are first principal components of household ownership.

6 Empirical Analysis

I estimate the effect of rainfall shocks on agricultural production using the following specification:

$$\text{asinh}(Y_{hdt}^c) = \sum_{s \in \{M, N\}} (\theta_{s-}^c R_{sht}^- + \theta_{s+}^c R_{sht}^+) + \sum_{s \in \{M, N\}} \gamma_s T_{sht} + X_{ht}^\top \delta + \alpha_h + \lambda_{dt} + \varepsilon_{hdt}^c \quad (6)$$

where $\text{asinh}(Y_{hdt}^c)$ is the inverse hyperbolic sine of agricultural production and is the quantity

harvested (in kilograms) of crop group c by household h in division d and round t estimated separately for total output, Aman (the monsoon, rainfed crop), and Boro (the dry-season, irrigated crop).⁵ θ_{s-}^c and θ_{s+}^c capture the effects of negative and positive rainfall shocks, respectively, in season $s \in \{\text{monsoon } M, \text{ non-monsoon } N\}$, where R_{sht}^- and R_{sht}^+ are the negative and positive share-of-normal deviations of rainfall from the household's long-run seasonal norm. T_{sht} are the corresponding standardised seasonal temperature deviations. X_{ht} is a vector of household controls, comprising household size and the numbers of elderly members and of children aged 0–4 and 5–17. α_h are household fixed effects, which absorb all time-invariant household characteristics, and λ_{dt} are wave-by-division fixed effects, which absorb shocks common to a division in a given round. Standard errors are clustered at the CHIRPS rainfall grid cell

I determine the impact of rainfall shocks on time allocation separately for men and women using the following specification:

$$Y_{ihdt}^a = \sum_{s \in \{M, N\}} (\beta_{s-}^a R_{sht}^- + \beta_{s+}^a R_{sht}^+) + \gamma' T_{sht} + X'_{iht} \delta + \alpha_i + \lambda_{dt} + \mu_m + \kappa_w + \varepsilon_{ihdt}^a \quad (7)$$

where Y_{ihdt}^a represents the number of minutes spent by individual i in household h in division d and round t on activity $a = 1, 2, 3, 4$ representing market, unpaid care, housework and leisure activities respectively. The β s capture the effect of rainfall shocks, both positive and negative, for monsoon and nonmonsoon periods. T_{ht} represents corresponding standardized temperature deviations. X'_{iht} is a vector of individual and household controls. α_i are individual fixed effects, λ_{dt} are wave-by-division fixed effects, and μ_m and κ_w are month-of-interview and day-of-week fixed effects respectively. Standard errors are clustered at the rainfall grid level.

⁵The inverse hyperbolic sine is used because it accommodates households with zero output for a given crop, which a logarithm cannot, while approximating a logarithm for larger quantities; the coefficients are therefore interpreted as approximate proportional effects.

7 Results

Table 3 reports the estimates from the impact of rainfall shocks on crop production. Column 1 reports the impact on total output, with no estimate being statistically significant. In Column 2, a 10 percentage point increase in both negative and positive monsoon shocks reduce the production of Aman rice by 27.5% and 19.1% respectively (both significant at the 10% level). Column 3 reports estimates from the impact of rainfall shocks on the dry season Boro crop. I find that the production of Boro crop is not impacted by rainfall shocks. As Aman rice crop is primarily rainfed, it operates through the income channel. On the other hand, Boro crop production relies primarily on irrigation and therefore is not impacted by rain.

Table 3: Rainfall Shocks and Agricultural Production

Variable	Total output	Aman (monsoon)	Boro (dry season)
Negative monsoon shock	-1.99 (1.28)	-2.75* (1.56)	0.35 (1.28)
Positive monsoon shock	-1.65 (1.01)	-1.91* (1.14)	-0.40 (0.95)
Negative non-monsoon shock	0.24 (1.70)	1.35 (1.90)	-0.77 (2.00)
Positive non-monsoon shock	-0.18 (0.18)	-0.09 (0.18)	-0.19 (0.15)
Observations	14,259	14,259	14,259
R-squared	0.807	0.747	0.786

Note: Standard errors, clustered at the rainfall grid level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 reports the estimates from the impact of rainfall shocks on time allocation. A 10-percentage point increase in negative monsoon shocks is associated with increasing unpaid care work for women by 7.2 minutes per day and decreasing time spent in leisure activities by 23.6 minutes per day. The estimates for both market and house work are not statistically significant. This provides evidence for an increase in the demand for caregiving. I find that a rainfall deficit experienced in the monsoon season increases the number of ill dependents by nearly 10%, with the incidence of fever among children under 5 increasing by 21% and

illness among the elderly rising by 17% (Table A6). A rainfall deficit in the monsoon season leads to reduced water availability, lower water quality and induces water storage leading to an increase in the incidence of illness including waterborne enteric fever such as typhoid and vector-borne fever such as dengue (Mogasale et al. 2018; Lowe et al. 2021; Dimitrova et al. 2022). These estimates also reveal that women face deepening time poverty and spend more time performing unpaid care work without a subsequent decrease in market work activities (Bardasi and Wodon 2006; Connolly 2018; Jiao, Li and Liu 2021; Torres-Higuera 2024; Boyd 2023; Garg, Gibson and Sun 2020).

A 10-percentage point increase in dry season rainfall deficit is associated with decreasing unpaid care work for women by 7.8 minutes per day, decreasing time spent in housework by 16.7 minutes per day and increasing time spent in leisure and other activities by 26.9 minutes per day. These findings are the exact opposite of what I find earlier in the case of monsoon shocks. In the monsoon season, negative rainfall shocks are associated with women spending more time in unpaid care yet in the dry season, they allocate less time to both unpaid care and housework. In the dry season, Boro crop is primarily irrigated and is therefore not responsive to rainfall shocks. Because of this, a dry season rainfall deficit does not reduce agricultural production. Furthermore, the nonmonsoon period precedes the monsoon shock and is thus measured nearly a year before responses in the time-use module. Because of this, I do not attribute a causal mechanism to the non-monsoon estimates. Instead, I interpret them as evidence that recent rainfall deficits increase women's time in caregiving but not during dry season rainfall deficits. A decline in time spent in domestic work by women by 15.7 minutes per day with time spent on cooking and shopping remaining unchanged does not reflect income-driven substitution between home and market production as this would also change time spent in cooking and shopping (Table A3). Women spend more time in leisure activities including more time spent in watching television or listening to radio (8.3 minutes per day) and in social activities (14.3 minutes per day) (Table A4).

For men, a 10 percentage point increase in positive rainfall shock is associated with a

fall in time spent in market work by 15 minutes per day. This is primarily driven by time spent in self-employment rather than in employed occupations or farm work (Table A5). Excess rainfall may lead to flooding and restrict mobility and lower local demand. On the other hand, the impact of rainfall shocks on men’s home production is marginal. In the case of nonmonsoon deficit rainfall, a 10 percentage point increase is associated with a 3.8 minute per day increase in unpaid care work. A 10 percentage point increase in excess nonmonsoon shocks increases time spent by men in housework by 2.3 minutes per day. These results underscore the gendered asymmetry inherent in response to rainfall shocks: while men adjust their time spent in market work activities due to being principal earners within the household, women are the primary providers of care and domestic work and are therefore more responsive to substituting their time between leisure and home production as a result of climate shocks (Emran and Shilpi 2018; Afridi, Mahajan and Sangwan 2022).

Table 4: Rainfall Shocks and Time Allocation

Variable	Men				Women			
	Market work	Unpaid care	Housework	Leisure	Market work	Unpaid care	Housework	Leisure
Negative monsoon shock	182.72 (123.20)	21.43 (15.64)	-95.09 (74.26)	-146.56 (111.90)	88.76 (59.31)	72.36** (36.46)	77.67 (70.98)	-236.18*** (76.59)
Positive monsoon shock	-149.60* (89.93)	3.14 (11.67)	17.91 (60.52)	127.44 (78.45)	25.91 (39.28)	-3.71 (28.81)	2.41 (51.84)	-23.91 (50.54)
Negative non-monsoon shock	-216.88 (140.16)	37.75** (17.82)	76.61 (95.93)	120.73 (124.97)	-27.32 (62.09)	-78.09* (41.71)	-167.49** (76.94)	268.60*** (87.64)
Positive non-monsoon shock	-7.11 (15.67)	-1.28 (1.99)	23.25** (9.79)	-18.73 (14.24)	-6.69 (7.53)	-6.80 (5.34)	12.19 (9.15)	-2.70 (10.57)
Observations	10,001	10,001	10,001	10,001	15,175	15,175	15,175	15,175
R-squared	0.592	0.427	0.513	0.606	0.547	0.498	0.527	0.524

Note: Standard errors, clustered at the CHIRPS 0.05-degree grid cell, in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

8 Conclusion

This paper examines the impact of rainfall shocks on time allocation in rural Bangladesh. I find that a 10 percentage point increase in negative monsoon shocks is associated with an increase in women’s unpaid care work and a decrease in the time she spends in leisure and other activities. This is due to increased demand for care: a monsoon rainfall deficit increases the incidence of fever among children under 5 and illness among the elderly. On

the other hand, a dry season rainfall deficit reduces the time women spend in housework and instead increases their time spent in leisure activities. Because this shock is measured nearly a year before the survey, I do not attribute a causal mechanism to it. On the other hand, men's time allocation is largely unresponsive to rainfall shocks. The exception to this is positive monsoon shocks reducing time men spend in market work.

My findings are in line with earlier literature which also finds women's market supply being unresponsive to rainfall shocks (Connolly 2018). Furthermore, women absorb weather shocks through unpaid care work without a proportionate decrease in market work, thereby deepening time poverty (Boyd 2023; Jiao, Li and Liu 2021; Vitellozzi and Giannelli 2024). Men, on the other hand, adjust the time they spend in market work activities where positive rainfall shocks may lead to reduced mobility or lower local demand. While earlier literature examines the impact of rainfall shocks on agricultural production and labor supply including the number of days worked and wages earned (Afridi, Mahajan and Sangwan 2022; Maitra and Tagat 2024), it does not examine the impact on intra-household time allocation. This paper tries to answer how rainfall shocks in different seasons have a differential impact on time allocation. While rainfall deficit during the monsoon season depresses agricultural production and increases the time women spend in unpaid care activities, the same shock in the nonmonsoon season has no impact on agricultural output and increases the time women allocate to leisure and other activities.

It is also important to note the limitations of this paper. First, this analysis is based on three rounds of panel data. Inclusion of more waves would allow a more comprehensive analysis of the impact of both current and lagged climate shocks on intra-household labor reallocation. Furthermore, there may also be measurement error inherent in the 24-hour time recall method adopted for the time allocation module. Future research can investigate the underlying mechanisms that can help explain the impact of rainfall shocks on time use dynamics. With South Asia being one of the most climate-vulnerable regions of the world, it is imperative to understand how rural households allocate labor such that more gender

sensitive protection systems can be devised.

References

- Abiona, Olukorede. 2017. “Adverse Effects of Early Life Extreme Precipitation Shocks on Short-term Health and Adulthood Welfare Outcomes.” *Review of Development Economics* 21(4): 1229–1254. <https://doi.org/10.1111/rode.12310>
- Addati, Laura, Umberto Cattaneo, Valeria Esquivel, and Isabel Valarino. 2018. *Care Work and Care Jobs for the Future of Decent Work*. Geneva: International Labour Organization.
- Afridi, Farzana, Kanika Mahajan, and Nikita Sangwan. 2022. “The Gendered Effects of Droughts: Production Shocks and Labor Response in Agriculture.” *Labour Economics* 78 (October): 102227. <https://doi.org/10.1016/j.labeco.2022.102227>
- Agamile, Peter, and David Lawson. 2021. “Rainfall Shocks and Children’s School Attendance: Evidence from Uganda.” *Oxford Development Studies* 49(3): 291–309. <https://doi.org/10.1080/13600818.2021.1895979>
- Aguilar, Arturo, and Marta Vicarelli. 2022. “El Niño and Children: Medium-Term Effects of Early-Life Weather Shocks on Cognitive and Health Outcomes.” *World Development* 150 (February): 105690. <https://doi.org/10.1016/j.worlddev.2021.105690>
- Ahmed, Sarah, Md. Zahid Hasan, Montira J. Pongsiri, Md. Wahid Ahmed, and Sylvia Szabo. 2021. “Effect of Extreme Weather Events on Injury, Disability, and Death in Bangladesh.” *Climate and Development* 13(4): 306–317. <https://doi.org/10.1080/17565529.2020.1780188>
- Akresh, Richard, Emilie Bagby, Damien de Walque, and Harounan Kazianga. 2012. *Child Labor, Schooling, and Child Ability*. Policy Research Working Paper No. 5965. Washington, DC: World Bank.
- Al Mamun, Md. Abdullah, Sheikh Arafat Islam Nihad, Md. Abdur Rouf Sarkar, Md. Abdullah Aziz, Md. Abdul Qayum, Rokib Ahmed, Niaz Md Farhat Rahman, Md. Ismail Hossain,

and Md. Shahjahan Kabir. 2021. “Growth and Trend Analysis of Area, Production and Yield of Rice: A Scenario of Rice Security in Bangladesh.” *PLOS ONE* 16(12): e0261128. <https://doi.org/10.1371/journal.pone.0261128>

Anxo, Dominique, Letizia Mencarini, Ariane Pailhé, Anne Solaz, Maria Letizia Tanturri, and Lennart Flood. 2011. “Gender Differences in Time Use over the Life Course in France, Italy, Sweden, and the US.” *Feminist Economics* 17(3): 159–195. <https://doi.org/10.1080/13545701.2011.582822>

Apps, Patricia. 2004. “Gender, Time Use, and Models of the Household.” Policy Research Working Paper No. 3233. Washington, DC: World Bank.

Auffhammer, Maximilian, Solomon M. Hsiang, Wolfram Schlenker, and Adam Sobel. 2013. “Using Weather Data and Climate Model Output in Economic Analyses of Climate Change.” *Review of Environmental Economics and Policy* 7(2): 181–198. <https://doi.org/10.1093/reep/ret016>

Bardasi, Elena, and Quentin Wodon. 2006. “Measuring Time Poverty and Analyzing Its Determinants: Concepts and Application to Guinea.” *Economics Bulletin* 10(12): 1–7.

Barnor, Kodjo, and Kashi Kafle. 2025. “Dry Weather, Empty Desks? Rainfall Shocks and Child Education in Tanzania.” Preprint, Unknown. <https://doi.org/10.22004/AG.ECON.361160>

Beegle, Kathleen, Rajeev H. Dehejia, and Roberta Gatti. 2006. “Child Labor and Agricultural Shocks.” *Journal of Development Economics* 81(1): 80–96. <https://doi.org/10.1016/j.jdeveco.2005.05.003>

Bernal, Carolina, and Razvan Vlaicu. 2023. “Child Labor, Rainfall Shocks, and Financial Inclusion: Evidence from Rural Households.” Inter-American Development Bank. <https://doi.org/10.18235/0005058>

- Bohorquez-Penuela, Camilo, and Andrea Otero-Cortes. 2020. “Blame It on the Rain: The Effects of Weather Shocks on Formal Rural Employment in Colombia.” Banco de la República de Colombia. <https://doi.org/10.32468/dtseru.292>
- Boyd, Christopher M. 2023. “Rainfall, Mothers’ Time Use, and Child Nutrition: Evidence from Rural Uganda.” *Population and Environment* 45(3): 17.
- Branco, Danyelle, and José Féres. 2021. “Weather Shocks and Labor Allocation: Evidence from Rural Brazil.” *American Journal of Agricultural Economics* 103(4): 1359–1377. <https://doi.org/10.1111/ajae.12171>
- Brown, Casey, Robyn Meeks, Yonas Ghile, and Kenneth Hunu. 2013. “Is Water Security Necessary? An Empirical Analysis of the Effects of Climate Hazards on National-Level Economic Growth.” *Philosophical Transactions of the Royal Society A* 371(2002): 20120416. <https://doi.org/10.1098/rsta.2012.0416>
- Burke, Marshall, and Kyle Emerick. 2016. “Adaptation to Climate Change: Evidence from US Agriculture.” *American Economic Journal: Economic Policy* 8(3): 106–140. <https://doi.org/10.1257/pol.20130025>
- Cain, Mead, Syeda Rokeya Khanam, and Shamsun Nahar. 1979. “Class, Patriarchy, and Women’s Work in Bangladesh.” *Population and Development Review* 5(3): 405–438. <https://doi.org/10.2307/1972079>
- Canessa, Eugenia, and Gianna Claudia Giannelli. 2021. *Women’s Employment and Natural Shocks*. IZA Discussion Paper No. 14055. Bonn: Institute of Labor Economics.
- Carpena, Fenella. 2019. “How Do Droughts Impact Household Food Consumption and Nutritional Intake? A Study of Rural India.” *World Development* 122 (October): 349–369. <https://doi.org/10.1016/j.worlddev.2019.06.005>

Carrillo, Bladimir. 2020. “Early Rainfall Shocks and Later-Life Outcomes: Evidence from Colombia.” *The World Bank Economic Review* 34(1): 179–209. <https://doi.org/10.1093/wber/lhy014>

Charmes, Jacques. 2019. *The Unpaid Care Work and the Labour Market: An Analysis of Time Use Data Based on the Latest World Compilation of Time-Use Surveys*. Geneva: International Labour Organization.

Chuang, Yating. 2019. “Climate Variability, Rainfall Shocks, and Farmers’ Income Diversification in India.” *Economics Letters* 174 (January): 55–61. <https://doi.org/10.1016/j.econlet.2018.10.015>

Colmer, Jonathan. 2021. “Rainfall Variability, Child Labor, and Human Capital Accumulation in Rural Ethiopia.” *American Journal of Agricultural Economics* 103(3): 858–877. <https://doi.org/10.1111/ajae.12128>

Connolly, Marie. 2008. “Here Comes the Rain Again: Weather and the Intertemporal Substitution of Leisure.” *Journal of Labor Economics* 26(1): 73–100. <https://doi.org/10.1086/522067>

Connolly, Marie. 2018. “Climate Change and the Allocation of Time.” *IZA World of Labor* 2018: 432. <https://doi.org/10.15185/izawol.432>

Corno, Lucia, Nicole Hildebrandt, and Alessandra Voena. 2020. “Age of Marriage, Weather Shocks, and the Direction of Marriage Payments.” *Econometrica* 88(3): 879–915. <https://doi.org/10.3982/ECTA15505>

Cosaert, Sam, Adrián Nieto Castro, and Konstantinos Tatsiramos. 2023. “Temperature and the Timing of Work.” IZA Discussion Paper No. 16480. Bonn: Institute of Labor Economics.

Damania, Richard, Sébastien Desbureaux, and Esha Zaveri. 2020. “Does Rainfall Matter for Economic Growth? Evidence from Global Sub-National Data

- (1990–2014).” *Journal of Environmental Economics and Management* 102: 102335. <https://doi.org/10.1016/j.jeem.2020.102335>
- Dastagir, M. Rehan. 2015. “Modeling Recent Climate Change Induced Extreme Events in Bangladesh: A Review.” *Weather and Climate Extremes* 7: 49–60. <https://doi.org/10.1016/j.wace.2014.10.003>
- Dehejia, Rajeev H., and Roberta Gatti. 2005. “Child Labor: The Role of Financial Development and Income Variability across Countries.” *Economic Development and Cultural Change* 53(4): 913–932. <https://doi.org/10.1086/428714>
- Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken. 2014. “What Do We Learn from the Weather? The New Climate-Economy Literature.” *Journal of Economic Literature* 52(3): 740–798. <https://doi.org/10.1257/jel.52.3.740>
- Démurger, Sylvie, Martin Fournier, and Weiyong Yang. 2010. “Rural Households Decisions Towards Income Diversification: Evidence from a Township in Northern China.” *China Economic Review* 21: S32–S44.
- Deschênes, Olivier, and Michael Greenstone. 2007. “The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather.” *American Economic Review* 97(1): 354–385. <https://doi.org/10.1257/aer.97.1.354>
- Diallo, Sadou, and Henri Atangana Ondo. 2025. “Climate Shocks and Labor Market in Sub-Saharan Africa: Effects on Gender Disparities in Urban and Rural Areas.” *Journal of Social and Economic Development* 27(1): 203–225. <https://doi.org/10.1007/s40847-024-00331-x>
- Diffenbaugh, Noah S., Deepti Singh, and Justin S. Mankin. 2018. “Unprecedented Climate Events: Historical Changes, Aspirational Targets, and National Commitments.” *Science Advances* 4(2). <https://doi.org/10.1126/sciadv.aao3354>

- Dillon, A. 2012. “Child Labour and Schooling Responses to Production and Health Shocks in Northern Mali.” *Journal of African Economies* 22(2): 276–299. <https://doi.org/10.1093/jae/ejs025>
- Dimitrova, Anna. 2021. “Seasonal Droughts and the Risk of Childhood Undernutrition in Ethiopia.” *World Development* 141 (May): 105417. <https://doi.org/10.1016/j.worlddev.2021.105417>
- Dimitrova, Anna, Sara McElroy, Morgan Levy, Alexander Gershunov, and Tarik Benmarhnia. 2022. “Precipitation Variability and Risk of Infectious Disease in Children under 5 Years for 32 Countries: A Global Analysis Using Demographic and Health Survey Data.” *The Lancet Planetary Health* 6 (2): e147–e155. [https://doi.org/10.1016/S2542-5196\(21\)00325-9](https://doi.org/10.1016/S2542-5196(21)00325-9)
- Dumas, Christelle. 2020. “Productivity Shocks and Child Labor: The Role of Credit and Agricultural Labor Markets.” *Economic Development and Cultural Change* 68(3): 763–812. <https://doi.org/10.1086/701828>
- Emran, Shahe, and Forhad Shilpi. 2018. “Agricultural Productivity, Hired Labor, Wages, and Poverty: Evidence from Bangladesh.” *World Development* 109 (September): 470–482. <https://doi.org/10.1016/j.worlddev.2016.12.009>
- Falaris, Evangelos M. 2023. “Children’s School Attendance, Work, Health and Rainfall Shocks in Ethiopia.” *Applied Economics* 55(40): 4609–4624. <https://doi.org/10.1080/00036846.2022.2130147>
- Farris, Jarrad, Maria Porter, Songqing Jin, and Mywish K. Maredia. 2023. “Growing Pains: Timing of In Utero Rainfall Shocks and Child Growth in Rural Rwanda.” *Economic Development and Cultural Change* 71(2): 793–818. <https://doi.org/10.1086/714886>
- Ferrant, Gaëlle, Luca Maria Pesando, and Keiko Nowacka. 2014. *Unpaid Care Work: The Missing Link in the Analysis of Gender Gaps in Labour Outcomes*. OECD Publishing. <https://doi.org/10.1787/1f3fd03f-en>

- Flato, Martin, Raya Muttarak, and André Pelser. 2017. “Women, Weather, and Woes: The Triangular Dynamics of Female-Headed Households, Economic Vulnerability, and Climate Variability in South Africa.” *World Development* 90 (February): 41–62. <https://doi.org/10.1016/j.worlddev.2016.08.015>
- Floro, Maria S. 2026. “Climate Change, Care Provisioning, and Inequality: Transitioning towards a Sustainable Economy.” *Oxford Review of Economic Policy* 41 (3–4): 1020–32. <https://doi.org/10.1093/oxrep/graf032>
- Frame, David J., Suzanne M. Rosier, Ilan Noy, Luke J. Harrington, Trevor Carey-Smith, Sarah N. Sparrow, Dáithí A. Stone, and Samuel M. Dean. 2020. “Climate Change Attribution and the Economic Costs of Extreme Weather Events: A Study on Damages from Extreme Rainfall and Drought.” *Climatic Change* 162: 781–797.
- Frege, Ilona Auer, Vanessa Bliecke, Sarah Bradshaw, Dennis Dijkzeul, Carolin Funke, Philipp Kienzl, Katharina Kusters, et al. 2023. “World Risk Report 2023.” Berlin. https://weltrisikobericht.de/wp-content/uploads/2023/10/WRR_2023_english_online161023.pdf
- Funk, Chris, Pete Peterson, Martin Landsfeld, Diego Pedreros, James Verdin, Shraddhanand Shukla, Gregory Husak, et al. 2015. “The Climate Hazards Infrared Precipitation with Stations—A New Environmental Record for Monitoring Extremes.” *Scientific Data* 2(1): 150066. <https://doi.org/10.1038/sdata.2015.66>
- Gao, Jianfeng, and Bradford F. Mills. 2018. “Weather Shocks, Coping Strategies, and Consumption Dynamics in Rural Ethiopia.” *World Development* 101 (January): 268–283. <https://doi.org/10.1016/j.worlddev.2017.09.002>
- Garg, Teevrat, Matthew Gibson, and Fanglin Sun. 2020. “Extreme Temperatures and Time Use in China.” *Journal of Economic Behavior & Organization* 180: 309–324. <https://doi.org/10.1016/j.jebo.2020.10.003>

- Graff Zivin, Joshua, and Matthew Neidell. 2014. "Temperature and the Allocation of Time: Implications for Climate Change." *Journal of Labor Economics* 32(1): 1–26. <https://doi.org/10.1086/671766>
- Gray, Clark, and Valerie Mueller. 2012. "Drought and Population Mobility in Rural Ethiopia." *World Development* 40(1): 134–145. <https://doi.org/10.1016/j.worlddev.2011.05.023>
- Heady, Christopher. 2003. "The Effect of Child Labor on Learning Achievement." *World Development* 31(2): 385–398. [https://doi.org/10.1016/S0305-750X\(02\)00186-9](https://doi.org/10.1016/S0305-750X(02)00186-9)
- Huq, Saleemul, Mizan Khan, A. K. M. Saiful Islam, and Af-sara Binte Mirza. 2024. "Climate Change Impacts in Bangladesh." https://www.icccad.net/wp-content/uploads/2024/02/Bangladesh_Final_Covers_26-Jan-2024_ONLINE_compressed.pdf
- ILO. 2024. "Child Labour." International Labour Organization. January 28, 2024. <https://www.ilo.org/projects-and-partnerships/projects/child-labour>
- ILO and UNICEF. 2021. *Child Labour: Global Estimates 2020, Trends and the Road Forward*. Geneva: International Labour Office.
- Islam, A. K. M. Saiful, S. B. Murshed, M. S. A. Khan, and M. A. Hasan. 2014. *Impact of Climate Change on Heavy Rainfall in Bangladesh*. Dhaka: Bangladesh University of Engineering and Technology.
- Islam, Mir Rabiul, Valerie Ingham, John Hicks, and Ian Manock. 2017. "The Changing Role of Women in Resilience, Recovery and Economic Development at the Intersection of Recurrent Disaster: A Case Study from Sirajganj, Bangladesh." *Journal of Asian and African Studies* 52(1): 50–67. <https://doi.org/10.1177/0021909614560244>

- Jensen, Robert. 2000. "Agricultural Volatility and Investments in Children." *American Economic Review* 90(2): 399–404. <https://doi.org/10.1257/aer.90.2.399>
- Jiao, Yang, Yuanyuan Li, and Miaojie Liu. 2021. "Widening the Gap? Temperature and Time Allocation between Men and Women." *Applied Economics* 53(5): 595–627.
- Jonasson, Erik, and Steven Helfand. 2010. "How Important are Locational Characteristics for Rural Non-Agricultural Employment? Lessons from Brazil." *World Development* 38: 727–741.
- Jost, Christine, Florence Kyazze, Jesse Naab, Sharmind Neelormi, James Kinyangi, Robert Zougmore, Pramod Aggarwal, et al. 2016. "Understanding Gender Dimensions of Agriculture and Climate Change in Smallholder Farming Communities." *Climate and Development* 8(2): 133–144. <https://doi.org/10.1080/17565529.2015.1050978>
- Kabeer, Naila, Lopita Huq, and Muhammad Mahabub Rahaman. 2021. "Material Barriers, Cultural Boundaries: A Mixed-Methods Analysis of Gender and Labour Market Segmentation in Bangladesh." WIDER Working Paper 2021/69. Helsinki: UNU-WIDER. <https://doi.org/10.35188/UNU-WIDER/2021/008-4>
- Khan, Mohammad Saddam, and Nadia Jannah. 2023. "National Child Labour Survey Bangladesh NCLS 2022." Dhaka: Bangladesh Bureau of Statistics.
- Krüger, Jens J., and Michael Neugart. 2018. "Weather and Intertemporal Labor Supply: Results from German Time-Use Data." *LABOUR* 32(1): 112–140.
- Le, Kien, and My Nguyen. 2022. "The Impacts of Rainfall Shocks on Birth Weight in Vietnam." *Journal of Development Effectiveness* 14(2): 143–159. <https://doi.org/10.1080/19439342.2021.1986114>
- Lee, Yevyng Beliyon Haile, Greg Seymour, and Carlo Azzarri. 2021. "The Heat Never Bothered Me Anyway: Gender-specific Response of Agricultural Labor to Cli-

matic Shocks in Tanzania.” *Applied Economic Perspectives and Policy* 43(2): 732–749.
<https://doi.org/10.1002/aepp.13153>

Lowe, Rachel, Sophie A. Lee, Kathleen M. O’Reilly, Oliver J. Brady, Leonardo Bastos, Gabriel Carrasco-Escobar, Rafael de Castro Catão, Felipe J. Colón-González, Christovam Barcellos, Marília Sá Carvalho, Marta Blangiardo, Håvard Rue, and Antonio Gasparrini. 2021. “Combined Effects of Hydrometeorological Hazards and Urbanisation on Dengue Risk in Brazil: A Spatiotemporal Modelling Study.” *The Lancet Planetary Health* 5 (4): e209–e219. [https://doi.org/10.1016/S2542-5196\(20\)30292-8](https://doi.org/10.1016/S2542-5196(20)30292-8)

Maccini, Sharon, and Dean Yang. 2009. “Under the Weather: Health, Schooling, and Economic Consequences of Early-Life Rainfall.” *American Economic Review* 99(3): 1006–1026.
<https://doi.org/10.1257/aer.99.3.1006>

Maitra, Pushkar, and Anirudh Tagat. 2024. “Labor Supply Responses to Rainfall Shocks.” *Review of Development Economics* 28(3): 851–871. <https://doi.org/10.1111/rode.13079>

Marchetta, Francesca, David E. Sahn, and Luca Tiberti. 2019. “The Role of Weather on Schooling and Work of Young Adults in Madagascar.” *American Journal of Agricultural Economics* 101(4): 1203–1227. <https://doi.org/10.1093/ajae/aaz015>

McKee, Thomas B., Nolan J. Doesken, and John Kleist. 1993. “The Relationship of Drought Frequency and Duration to Time Scales.” In *Proceedings of the 8th Conference on Applied Climatology*, 179–184. Anaheim, CA: American Meteorological Society.

Mogasale, Vijayalaxmi V., Enusa Ramani, Vittal Mogasale, Ju Yeon Park, and Thomas F. Wierzba. 2018. “Estimating Typhoid Fever Risk Associated with Lack of Access to Safe Water: A Systematic Literature Review.” *Journal of Environmental and Public Health* 2018 (July): 9589208. <https://doi.org/10.1155/2018/9589208>

- Namhata, Anindita, Zakir Husain, YuJung Lee, and Mousumi Dutta. 2025. “Rain and Time Use in Bangladesh: Does Rain Reduce Gender Gaps?” Working Paper. Kolkata: Presidency University.
- Nasuda, Akiko, Hiromitsu Kanno, and Takeshi Sakurai. 2011. “Analysis of Impact of Heavy Rainfall Shocks on Time Allocation Changes in Rural Zambia.” Working Paper.
- Newman, Rebecca, and Ilan Noy. 2023. “The Global Costs of Extreme Weather That Are Attributable to Climate Change.” *Nature Communications* 14(1): 6103.
- Nguyen, Ha Trong, Huong Thu Le, and Luke B. Connelly. 2021. “Weather and Children’s Time Allocation.” *Health Economics* 30(7): 1559–1579.
- Patz, Jonathan A., Diarmid Campbell-Lendrum, Tracey Holloway, and Jonathan A. Foley. 2005. “Impact of Regional Climate Change on Human Health.” *Nature* 438(7066): 310–317. <https://doi.org/10.1038/nature04188>
- Pham, Trinh. 2021. “Adverse Rainfall Shocks and Child Outcomes: Evidence from Rural Vietnam.” <https://doi.org/10.22004/ag.econ.314051>
- Ponnusamy, Sundar. 2026. “The Impact of Rainfall Shocks in Childhood and Adolescence on Human Capital Development.” *American Journal of Agricultural Economics* 108(1): 383–405. <https://doi.org/10.1111/ajae.12547>
- Rao, Nitya, Elaine T. Lawson, Wapula N. Raditloaneng, Divya Solomon, and Margaret N. Angula. 2019. “Gendered Vulnerabilities to Climate Change: Insights from the Semi-Arid Regions of Africa and Asia.” *Climate and Development* 11(1): 14–26. <https://doi.org/10.1080/17565529.2017.1372266>
- Rocha, Rudi, and Rodrigo R. Soares. 2015. “Water Scarcity and Birth Outcomes in the Brazilian Semi-arid.” *Journal of Development Economics* 112 (January): 72–91. <https://doi.org/10.1016/j.jdeveco.2014.10.003>

Rosales-Rueda, Maria. 2018. “The Impact of Early Life Shocks on Human Capital Formation: Evidence from El Niño Floods in Ecuador.” *Journal of Health Economics* 62 (November): 13–44. <https://doi.org/10.1016/j.jhealeco.2018.07.003>

Rose, Elaina. 2001. “Ex Ante and Ex Post Labor Supply Response to Risk in a Low-Income Area.” *Journal of Development Economics* 64(2): 371–388. [https://doi.org/10.1016/S0304-3878\(00\)00142-5](https://doi.org/10.1016/S0304-3878(00)00142-5)

Sayer, Liana C. 2005. “Gender, Time and Inequality: Trends in Women’s and Men’s Paid Work, Unpaid Work and Free Time.” *Social Forces* 84(1): 285–303. <https://doi.org/10.1353/sof.2005.0126>

Schiermeier, Quirin. 2018. “Droughts, Heatwaves and Floods: How to Tell When Climate Change Is to Blame.” *Nature* 560(7717): 20–23.

Sesmero, Juan, Jacob Ricker-Gilbert, and Aaron Cook. 2018. “How Do African Farm Households Respond to Changes in Current and Past Weather Patterns? A Structural Panel Data Analysis from Malawi.” *American Journal of Agricultural Economics* 100(1): 115–144. <https://doi.org/10.1093/ajae/aax068>

Shah, Manisha, and Bryce Millett Steinberg. 2017. “Drought of Opportunities: Contemporaneous and Long-Term Impacts of Rainfall Shocks on Human Capital.” *Journal of Political Economy* 125(2): 527–561. <https://doi.org/10.1086/691788>

Solotaroff, Jennifer L., Afrah Kotikula, Tara Lonnberg, Satu Ali, and Ferdous Jahan. 2019. *Voices to Choices: Bangladesh’s Journey in Women’s Economic Empowerment*. Washington, DC: World Bank.

Somanathan, E., Rohini Somanathan, Anant Sudarshan, and Meenu Tewari. 2021. “The Impact of Temperature on Productivity and Labor Supply: Evidence from Indian Manufacturing.” *Journal of Political Economy* 129(6): 1797–1827. <https://doi.org/10.1086/713733>

- Staffieri, Irene, Nicholas J. Sitko, and John A. Maluccio. 2023. “Sustaining Enrolment When Rains Fail: School Feeding, Rainfall Shocks and Schooling in Malawi.” *Food Policy* 121 (November): 102539. <https://doi.org/10.1016/j.foodpol.2023.102539>
- Thai, Thuan Q., and Evangelos M. Falaris. 2014. “Child Schooling, Child Health, and Rainfall Shocks: Evidence from Rural Vietnam.” *The Journal of Development Studies* 50(7): 1025–1037. <https://doi.org/10.1080/00220388.2014.903247>
- Tiwari, Sailesh, Hanan G. Jacoby, and Emmanuel Skoufias. 2017. “Monsoon Babies: Rainfall Shocks and Child Nutrition in Nepal.” *Economic Development and Cultural Change* 65(2): 167–188. <https://doi.org/10.1086/689308>
- Torres-Higuera, Paula. 2024. “Warm Days, Warmer Homes? Effects of Temperature Shocks on Time Allocation.” Documento CEDE No. 15. Bogotá: Universidad de los Andes. <https://ssrn.com/abstract=4816721>
- Trinh, Trong-Anh, Alberto Posso, and Simon Feeny. 2020. “Child Labor and Rainfall Deviation: Panel Data Evidence from Rural Vietnam.” *The Developing Economies* 58(1): 63–76. <https://doi.org/10.1111/deve.12215>
- Trinh, Trong-Anh, and Quanda Zhang. 2021. “Adverse Shocks, Household Expenditure and Child Marriage: Evidence from India and Vietnam.” *Empirical Economics* 61(3): 1617–1639. <https://doi.org/10.1007/s00181-020-01907-2>
- Vitellozzi, Sveva, and Gianna Claudia Giannelli. 2024. “Thriving in the Rain: Natural Shocks, Time Allocation, and Women’s Empowerment in Bangladesh.” *World Development* 181 (September): 106684. <https://doi.org/10.1016/j.worlddev.2024.106684>
- Vogel, Elisabeth, Markus G. Donat, Lisa V. Alexander, et al. 2019. “The Effects of Climate Extremes on Global Agricultural Yields.” *Environmental Research Letters* 14(5): 054010. <https://doi.org/10.1088/1748-9326/ab154b>

Yohannes, Dereje, Annika Lindskog, and Assefa Admassie. 2026. “Effects of Rainfall Shocks on Child Labour: Panel Data Evidence from Ethiopia.” *Journal of African Economies* 35(1): ejaf014. <https://doi.org/10.1093/jae/ejaf014>

Zohir, Sajjad, Susmita Dutta, Siddiqur Rahman, and Wasama Ahmed Khan. 2024. “Child Labour and Schooling in Bangladesh: Insights from Data and Evidence.” UNICEF.

A Additional Tables and Figures

Table A.1: Components of the Asset Indexes

Durable assets (D1)	Agricultural assets (D2)	Livestock (K1)
Trunk/suitcase	Kaste	Bullocks
Cooking pots	Nirani	Milk cows
Bed, cabinet, table	Ladder, rake	Buffalo
Electric fan, iron	Plough and yoke	Goats
Radio, cassette player	Manual sprayer	Sheep
Clock, wristwatch	Wheelbarrow, carts	Chickens
Television	Tractor	Layer chickens
Jewellery	Power tiller	Broiler chickens
Sewing machine	Thresher, harvester	Cockerels
Bicycle, rickshaw, van	Fodder cutter	Ducks
Boat, engine boat	Swing basket, don	Other birds
Motorcycle	Hand tubewell	Pigs
Mobile, landline phone	Treadle/rower pump	Other livestock
Dheki, jata, randa	Low-lift pump	
Saw, hammer, axe	Shallow tubewell	
Fishing net, spade	Deep tubewell	
Solar panel, generator	Electric/diesel pump	
Printer, tablet, camera	Mason's equipment	
Draught animals	Potter's wheel	

Table A.2: Rainfall Shocks and Time Allocation: Full Coefficients

Variable	Men				Women			
	Market work	Unpaid care	Housework	Leisure	Market work	Unpaid care	Housework	Leisure
Negative monsoon shock	182.72 (123.20)	21.43 (15.64)	-95.09 (74.26)	-146.56 (111.90)	88.76 (59.31)	72.36** (36.46)	77.67 (70.98)	-236.18*** (76.59)
Positive monsoon shock	-149.60* (89.93)	3.14 (11.67)	17.91 (60.52)	127.44 (78.45)	25.91 (39.28)	-3.71 (28.81)	2.41 (51.84)	-23.91 (50.54)
Negative non-monsoon shock	-216.88 (140.16)	37.75** (17.82)	76.61 (95.93)	120.73 (124.97)	-27.32 (62.09)	-78.09* (41.71)	-167.49** (76.94)	268.60*** (87.64)
Positive non-monsoon shock	-7.11 (15.67)	-1.28 (1.99)	23.25** (9.79)	-18.73 (14.24)	-6.69 (7.53)	-6.80 (5.34)	12.19 (9.15)	-2.70 (10.57)
Temperature shock (monsoon)	8.46 (27.26)	-2.21 (3.15)	-45.56*** (15.96)	38.74* (23.42)	-16.17 (11.47)	-12.83* (7.66)	12.36 (15.52)	12.82 (15.42)
Temperature shock (non-monsoon)	-2.70 (22.30)	2.21 (2.82)	19.73 (13.87)	-18.65 (20.71)	14.96 (10.10)	-3.99 (6.71)	19.00 (12.85)	-29.50** (12.82)
Household size	2.91 (5.67)	-0.57 (0.75)	-3.67 (3.59)	1.08 (5.21)	1.80 (2.27)	1.13 (1.71)	-6.86** (3.17)	3.94 (3.18)
Number of elderly (60+)	-18.44* (10.85)	1.24 (1.34)	2.93 (7.02)	13.29 (9.84)	-8.25* (4.61)	4.50 (3.07)	2.31 (6.07)	2.48 (6.41)
Number of children aged 0–4	0.66 (9.99)	6.94*** (1.48)	6.37 (5.80)	-12.94 (8.93)	-11.76*** (4.01)	42.48*** (3.27)	-7.64 (5.42)	-25.04*** (5.90)
Number of children aged 5–17	2.98 (6.60)	0.21 (0.91)	5.62 (4.23)	-8.34 (6.17)	-0.72 (2.82)	5.16** (2.16)	9.96** (3.95)	-15.46*** (4.11)
Durable asset index	2.14 (3.64)	-0.18 (0.53)	-2.02 (2.22)	-0.22 (3.17)	2.81* (1.65)	0.24 (1.05)	-1.23 (1.93)	-1.09 (2.17)
Agricultural asset index	3.00 (4.04)	0.08 (0.48)	0.40 (2.36)	-3.40 (3.71)	4.54*** (1.73)	-0.72 (1.17)	1.25 (2.37)	-4.82** (2.40)
Livestock index	9.58** (3.92)	-0.83 (0.51)	4.10 (2.72)	-12.86*** (4.36)	9.23*** (2.23)	-1.97* (1.16)	6.26** (2.54)	-14.87*** (3.21)
Observations	10,001	10,001	10,001	10,001	15,175	15,175	15,175	15,175
R-squared	0.592	0.427	0.513	0.606	0.547	0.498	0.527	0.524

Note: Full coefficients from the specification reported in Table 4. OLS, men and women estimated separately. Outcomes are minutes per day. All columns include individual, wave-by-division, day-of-week and month-of-interview fixed effects. Standard errors, clustered at the CHIRPS 0.05-degree grid cell, in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A.3: Women: Housework and Care Activities

Variable	Shopping	Cooking	Domestic work	Unpaid care
Negative monsoon shock	2.78 (7.37)	-0.34 (32.13)	75.24 (61.18)	72.36** (36.46)
Positive monsoon shock	-4.35 (5.37)	11.38 (24.17)	-4.62 (43.96)	-3.71 (28.81)
Negative non-monsoon shock	-5.10 (7.93)	-5.39 (35.42)	-157.00** (66.51)	-78.09* (41.71)
Positive non-monsoon shock	0.56 (1.34)	-1.25 (4.67)	12.88 (8.44)	-6.80 (5.34)
Observations	15,175	15,175	15,175	15,175
R-squared	0.386	0.484	0.495	0.498

Note: Activity-level decomposition of the housework and unpaid-care categories in Table 4, for women. Outcomes in minutes per day. All columns include individual, wave-by-division, day-of-week and month-of-interview fixed effects and the estimation controls. Standard errors, clustered at the CHIRPS 0.05-degree grid cell, in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A.4: Women: Leisure Activities

Variable	TV/radio	Exercising	Social	Religious	Other
Negative monsoon shock	1.84 (20.92)	6.31 (6.71)	-83.47* (43.23)	8.70 (31.47)	1.81 (10.78)
Positive monsoon shock	-5.95 (16.65)	-2.72 (4.61)	-51.54 (39.15)	13.27 (20.15)	6.37 (9.96)
Negative non-monsoon shock	83.32*** (27.22)	-2.18 (8.47)	142.97** (55.55)	17.30 (31.63)	-10.59 (14.50)
Positive non-monsoon shock	-5.71* (3.27)	0.25 (1.10)	7.29 (6.79)	-2.88 (5.20)	-0.85 (1.65)
Observations	15,175	15,175	15,175	15,175	15,175
R-squared	0.561	0.428	0.534	0.601	0.386

Note: Activity-level decomposition of the leisure category in Table 4, for women. Outcomes in minutes per day. All columns include individual, wave-by-division, day-of-week and month-of-interview fixed effects and the estimation controls. Standard errors, clustered at the CHIRPS 0.05-degree grid cell, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Men: Market-Work Activities

Variable	Employed	Own business	Farming/fishing	Weaving
Negative monsoon shock	228.18** (97.67)	-131.16 (96.02)	73.01 (85.97)	12.69* (7.60)
Positive monsoon shock	18.11 (78.91)	-180.23*** (67.93)	7.73 (70.31)	4.80 (7.94)
Negative non-monsoon shock	-104.15 (122.73)	-54.17 (119.22)	-46.31 (105.62)	-12.24 (11.98)
Positive non-monsoon shock	-4.73 (13.65)	-10.05 (13.40)	7.68 (11.89)	-0.02 (0.72)
Observations	10,001	10,001	10,001	10,001
R-squared	0.587	0.667	0.607	0.459

Note: Activity-level decomposition of the market-work category in Table 4, for men. Outcomes in minutes per day. All columns include individual, wave-by-division, day-of-week and month-of-interview fixed effects and the estimation controls. Standard errors, clustered at the CHIRPS 0.05-degree grid cell, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Rainfall Shocks and Household Illness

	(1) Ill dependents	(2) Fever, under-5	(3) Illness, elderly 60+
Negative monsoon shock	0.764*** (0.266)	0.337*** (0.112)	0.362*** (0.103)
Positive monsoon shock	−0.335* (0.182)	−0.028 (0.083)	−0.120 (0.075)
Negative non-monsoon shock	−1.282*** (0.341)	−0.074 (0.130)	−0.423*** (0.122)
Positive non-monsoon shock	−0.162*** (0.042)	−0.054*** (0.016)	−0.006 (0.015)
Temperature controls	Yes	Yes	Yes
Household controls	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Wave × Division FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes
Month-of-interview FE	Yes	Yes	Yes
Observations	15,169	15,169	15,169
Mean of dep. var.	0.723	0.159	0.214

Notes: Each column reports a separate OLS regression on the sample of female time-use respondents observed in at least two rounds. The dependent variable is the number of household members with the indicated condition, reported ill in the four weeks before the survey (Module W4): column (1) counts non-working-age dependents (children under 15 or adults 60+) with any illness; (2) children under 5 with fever; (3) members aged 60+ with any illness. Coefficients are per unit of the season's mean weekly share-of-normal rainfall deviation; a ten-percentage-point change corresponds to the coefficient divided by ten. Household controls are household size and the numbers of elderly, children aged 0–4 and 5–17, and durable, agricultural and livestock asset indices. Standard errors clustered at the CHIRPS grid cell in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

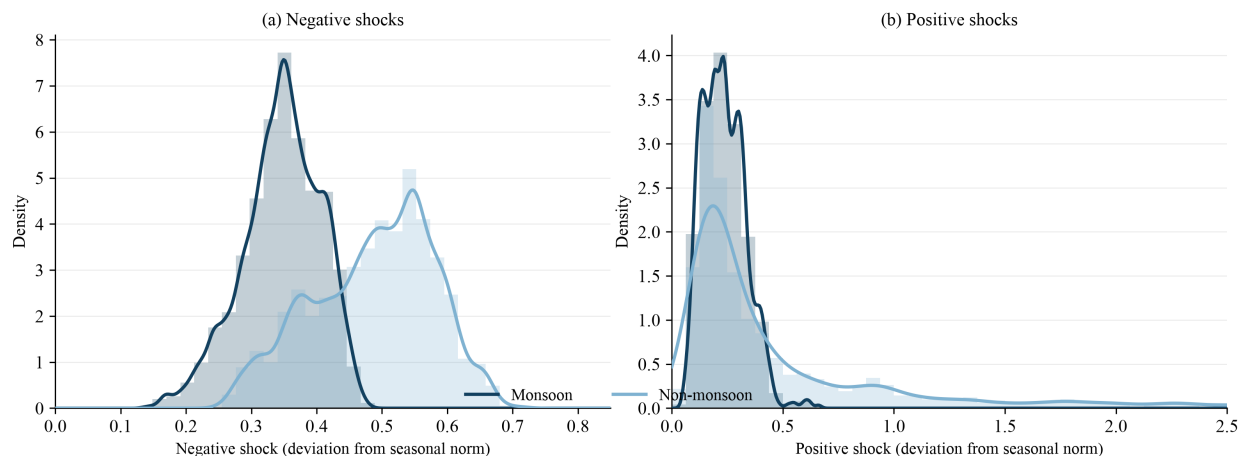


Figure A.1: Distribution of current rainfall shocks by season.

Notes: Kernel densities (with underlying histograms) of the household-level current rainfall shocks, measured as the average seasonal deviation from the 1981–2011 norm, for the monsoon (Jun–Oct) and non-monsoon (Nov–May) seasons. Panel (a) shows negative shocks (rainfall deficits) and panel (b) positive shocks (rainfall surpluses). Household-by-wave level ($N = 16,185$); the positive non-monsoon axis is truncated at 2.5 (maximum ≈ 3.5).

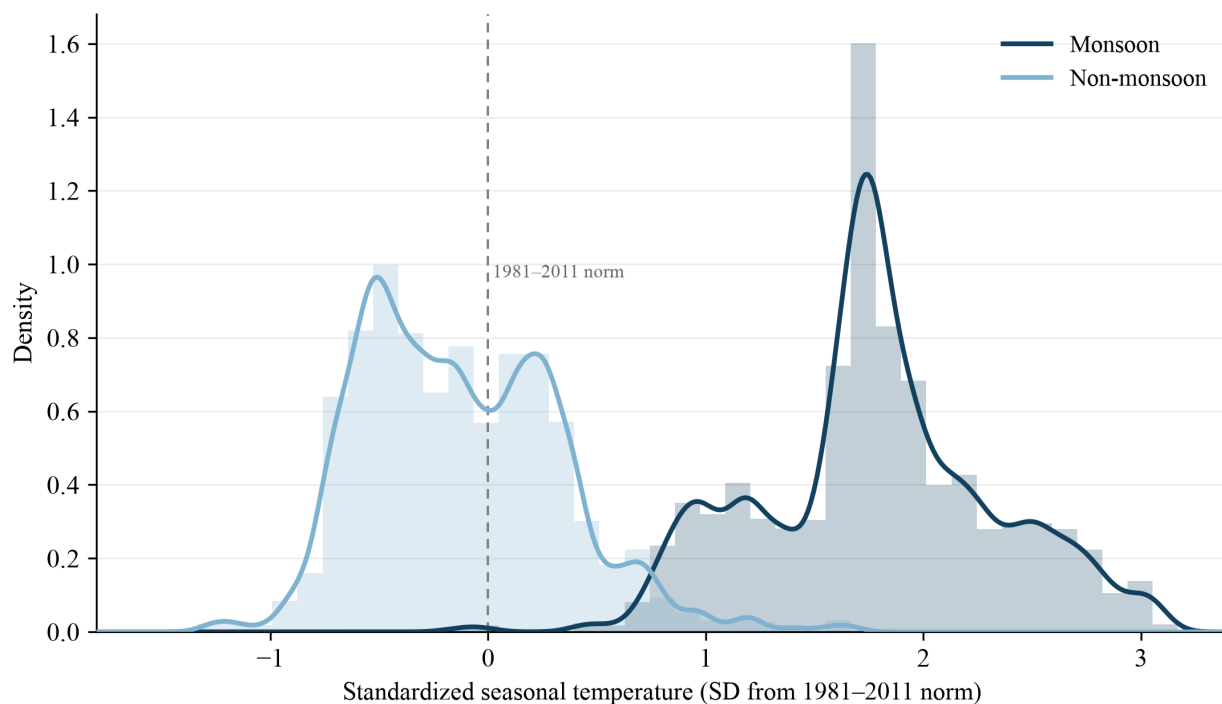


Figure A.2: Weekly mean temperature in Bangladesh, 2010–2019.

Notes: Weekly mean 2-m air temperature computed from ERA5 hourly data, averaged over all grid cells covering Bangladesh and over the days in each week of the year. Thin lines denote individual years; the bold line is the 2010–2019 mean. The shaded band marks the monsoon season (June–October).

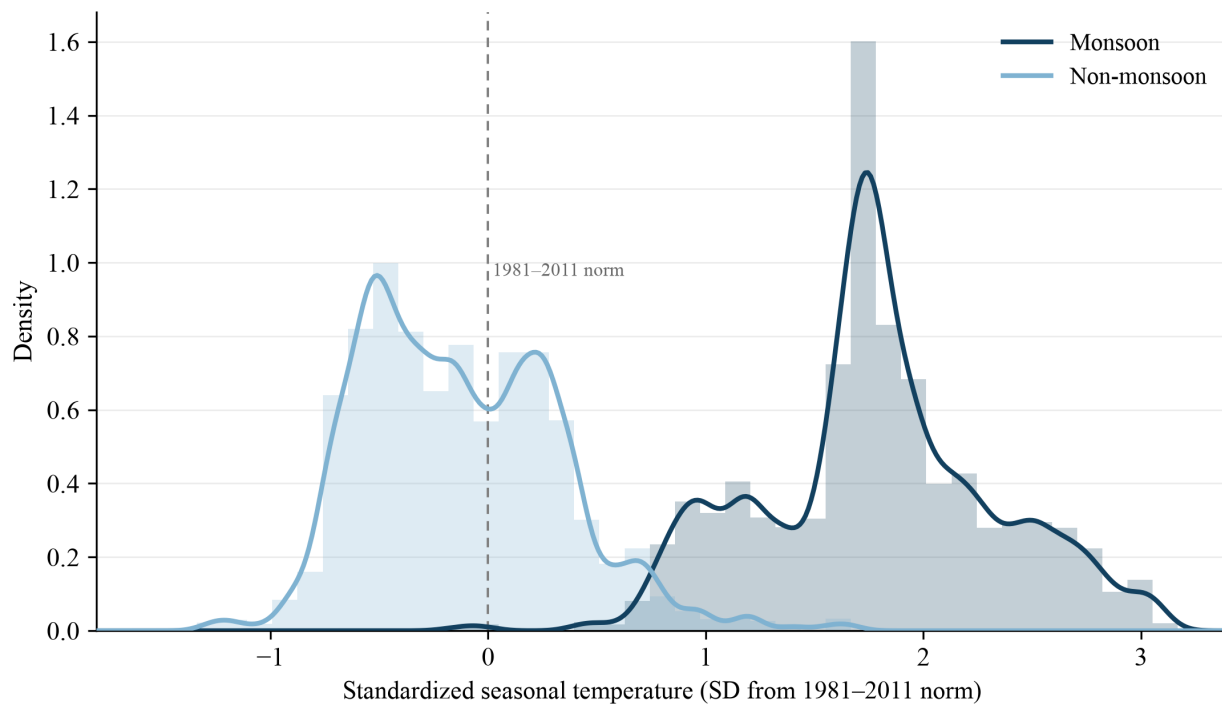


Figure A.3: Distribution of standardized seasonal temperature by season.

Notes: Kernel densities (with underlying histograms) of the household-level standardized seasonal temperature, measured in standard deviations from the 1981–2011 seasonal norm, for the monsoon (Jun–Oct) and non-monsoon (Nov–May) seasons. Values above zero are warmer than the historical norm. Household-by-wave level ($N = 16,185$).

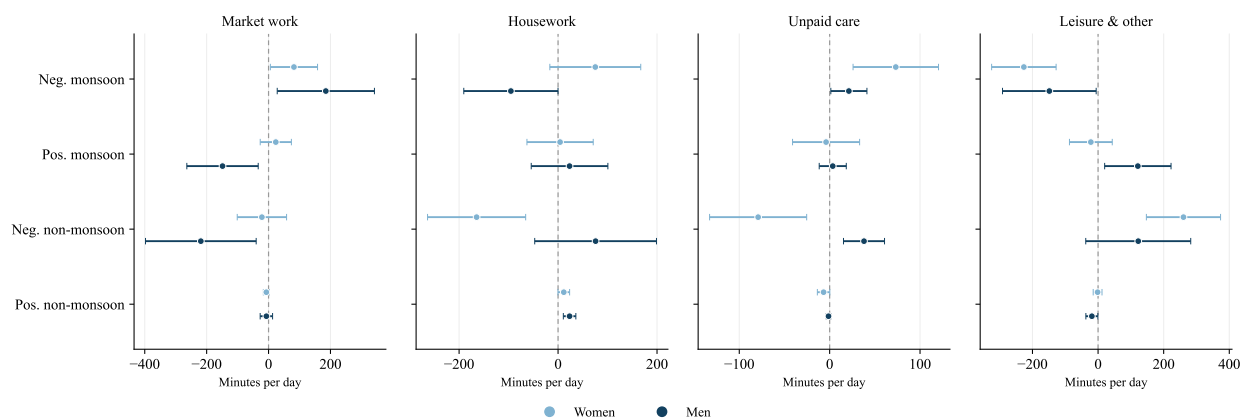


Figure A.4: Rainfall shocks and time allocation, by sex.

Notes: Coefficients (minutes/day) of rainfall shocks on each time-use outcome, estimated separately for women and men; standard errors clustered at the rainfall grid. Bars are 90% confidence intervals.

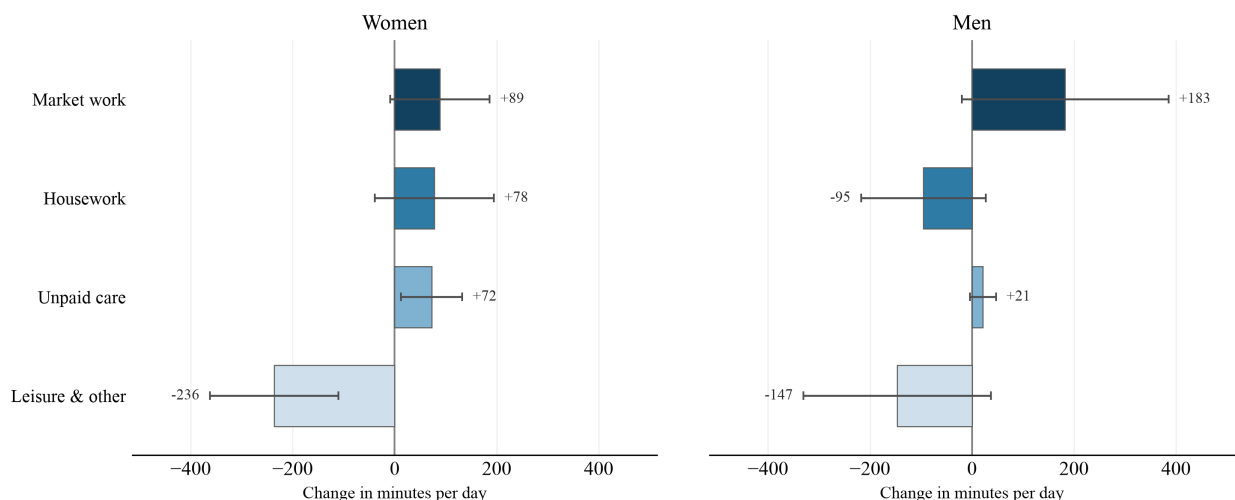


Figure A.5: Reallocation of the 24-hour day under a negative monsoon rainfall shock.
Notes: Bars are the estimated change in daily minutes in each activity for a one-unit negative monsoon rainfall shock, from the individual-FE OLS (whiskers: 90% confidence intervals, standard errors clustered at the rainfall grid).

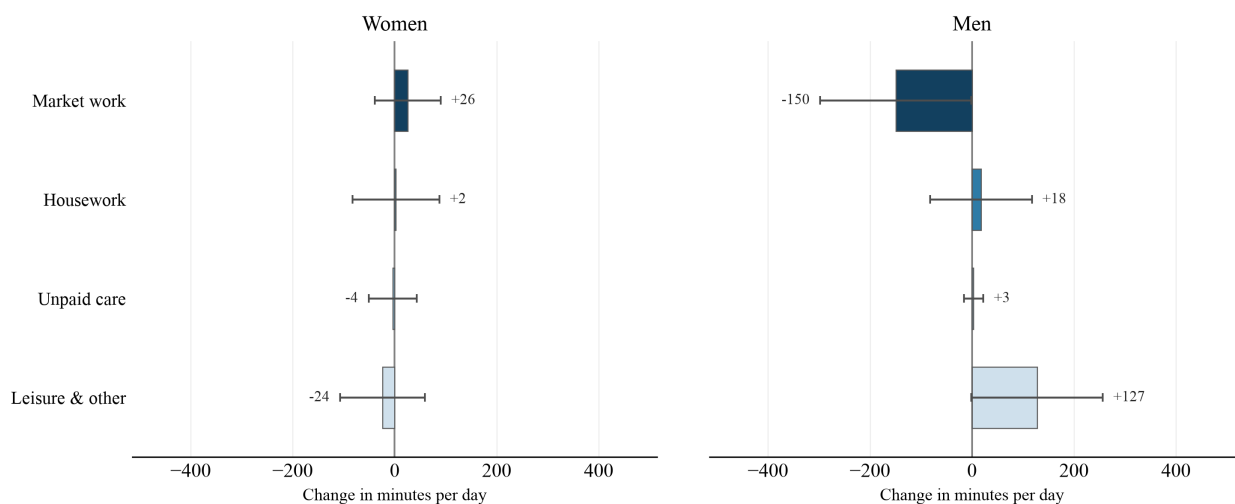


Figure A.6: Reallocation of the 24-hour day under a positive monsoon rainfall shock.
Notes: Bars are the estimated change in daily minutes in each activity for a one-unit positive monsoon rainfall shock, from the individual-FE OLS (whiskers: 90% confidence intervals, standard errors clustered at the rainfall grid).

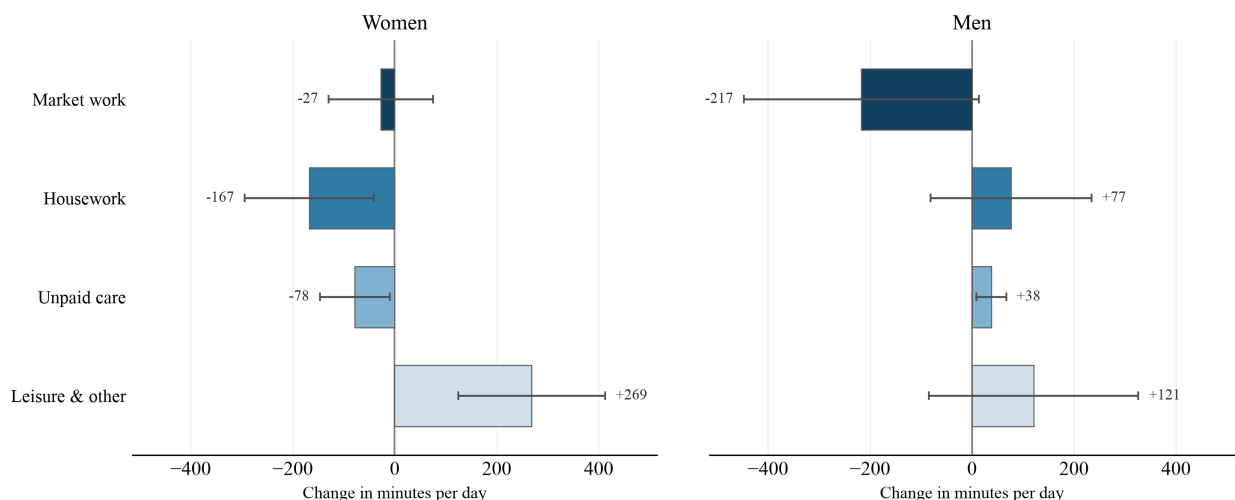


Figure A.7: Reallocation of the 24-hour day under a negative non-monsoon rainfall shock. *Notes:* Bars are the estimated change in daily minutes in each activity for a one-unit negative non-monsoon rainfall shock, from the individual-FE OLS (whiskers: 90% confidence intervals, standard errors clustered at the rainfall grid).

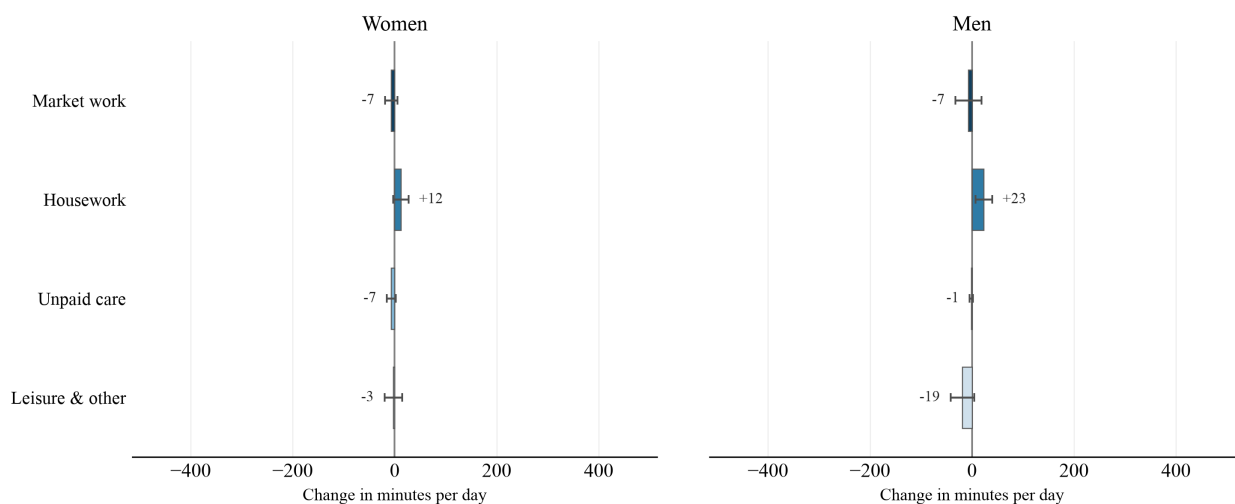


Figure A.8: Reallocation of the 24-hour day under a positive non-monsoon rainfall shock. *Notes:* Bars are the estimated change in daily minutes in each activity for a one-unit positive non-monsoon rainfall shock, from the individual-FE OLS (whiskers: 90% confidence intervals, standard errors clustered at the rainfall grid).